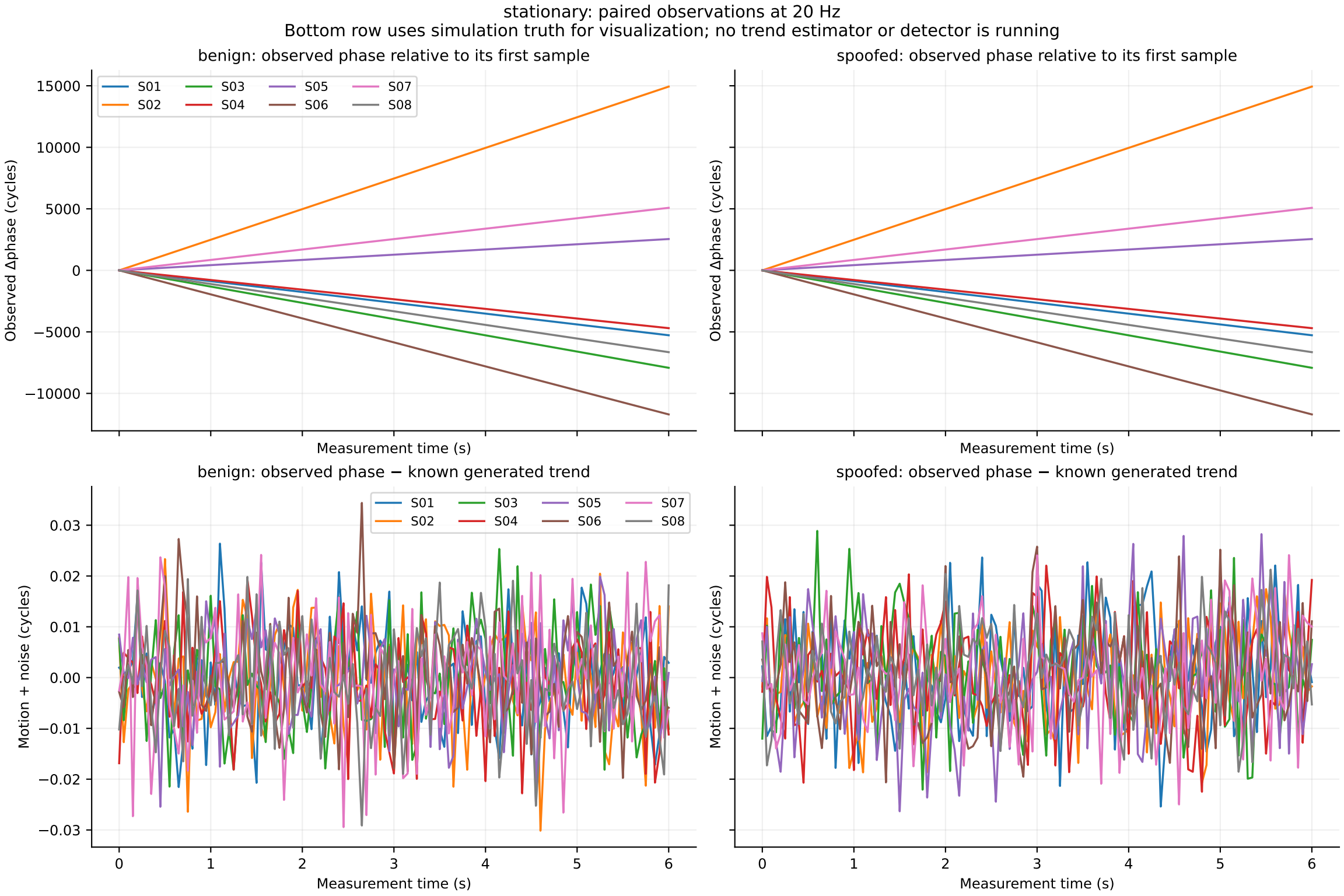
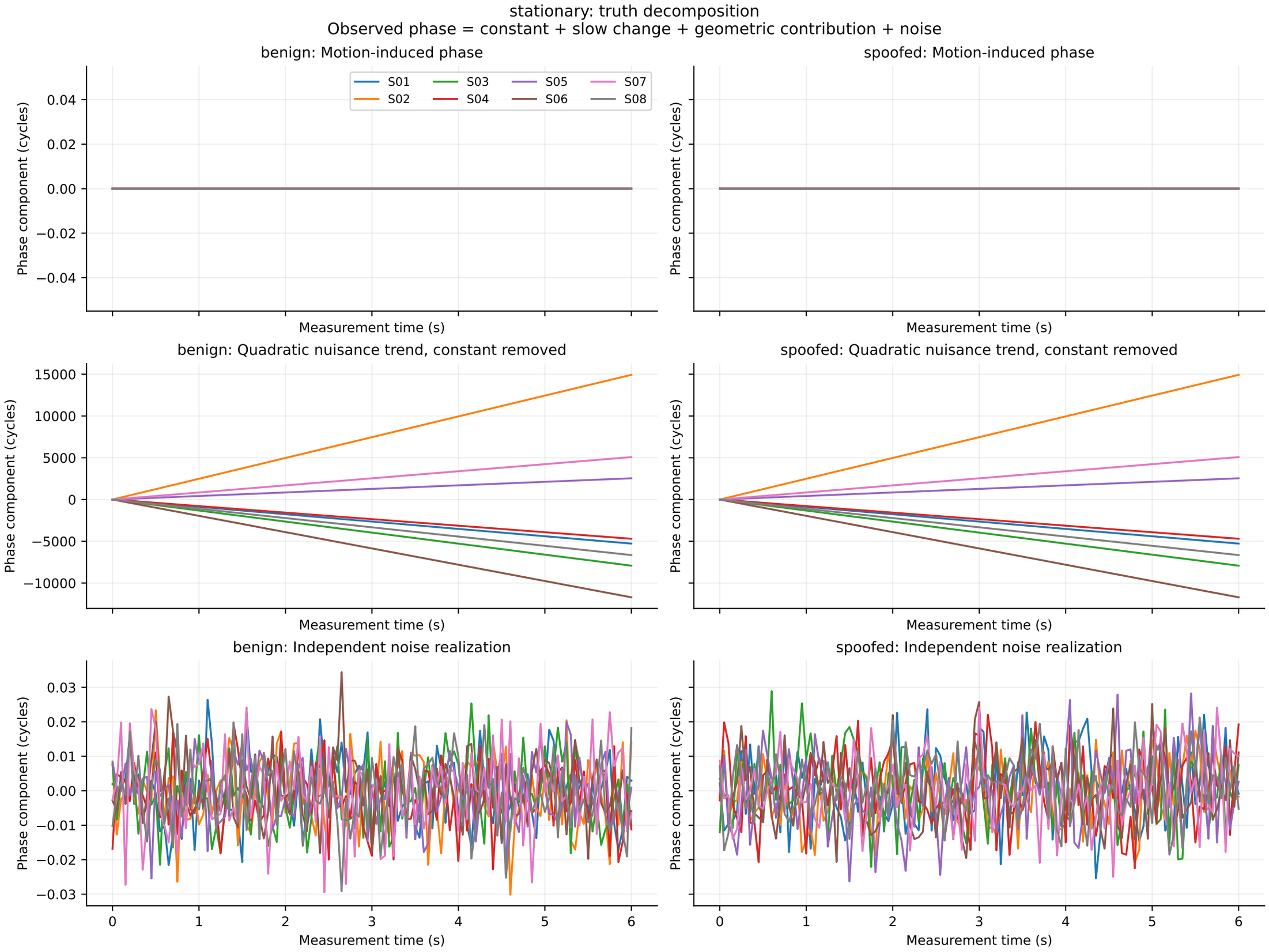
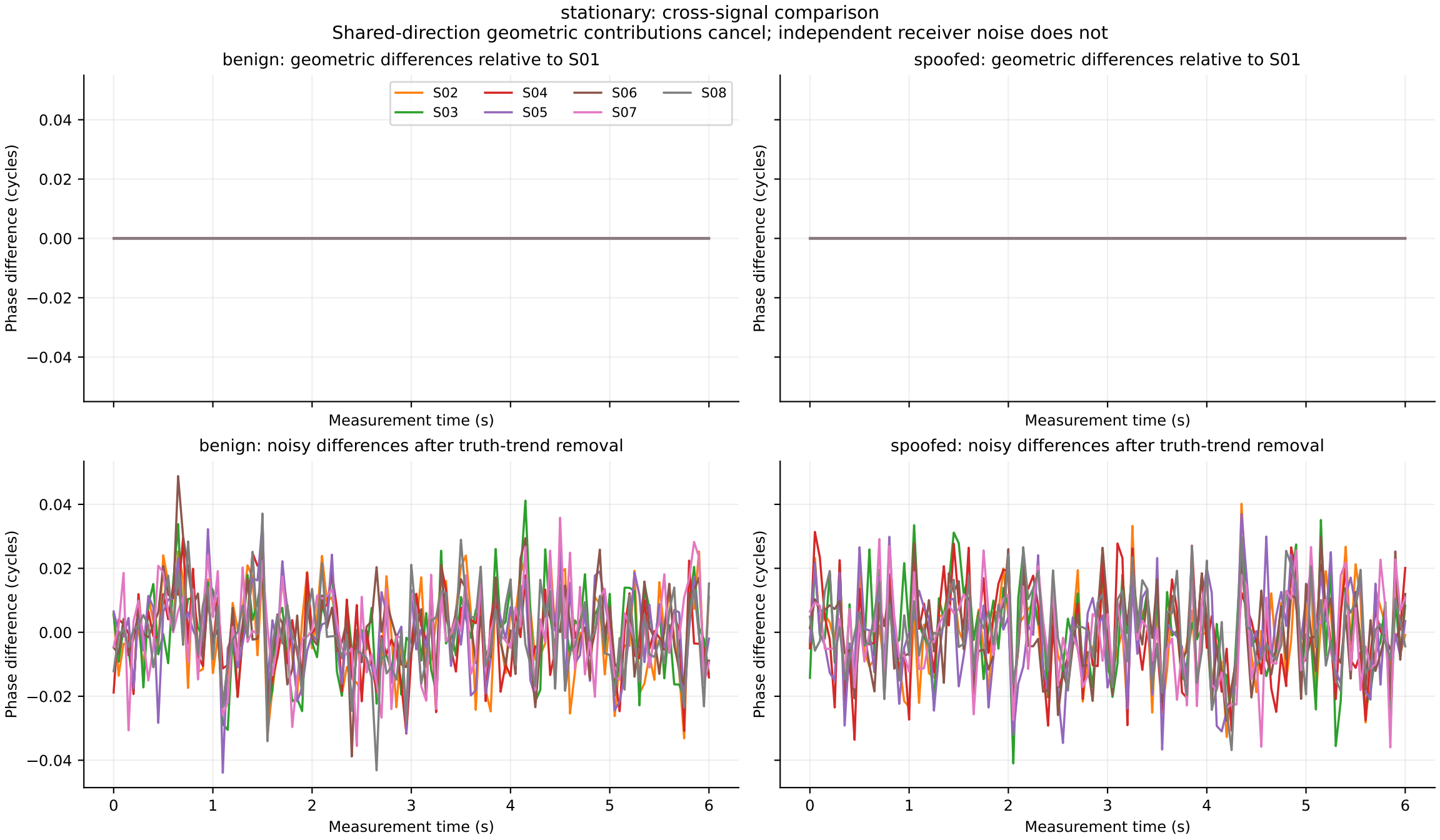
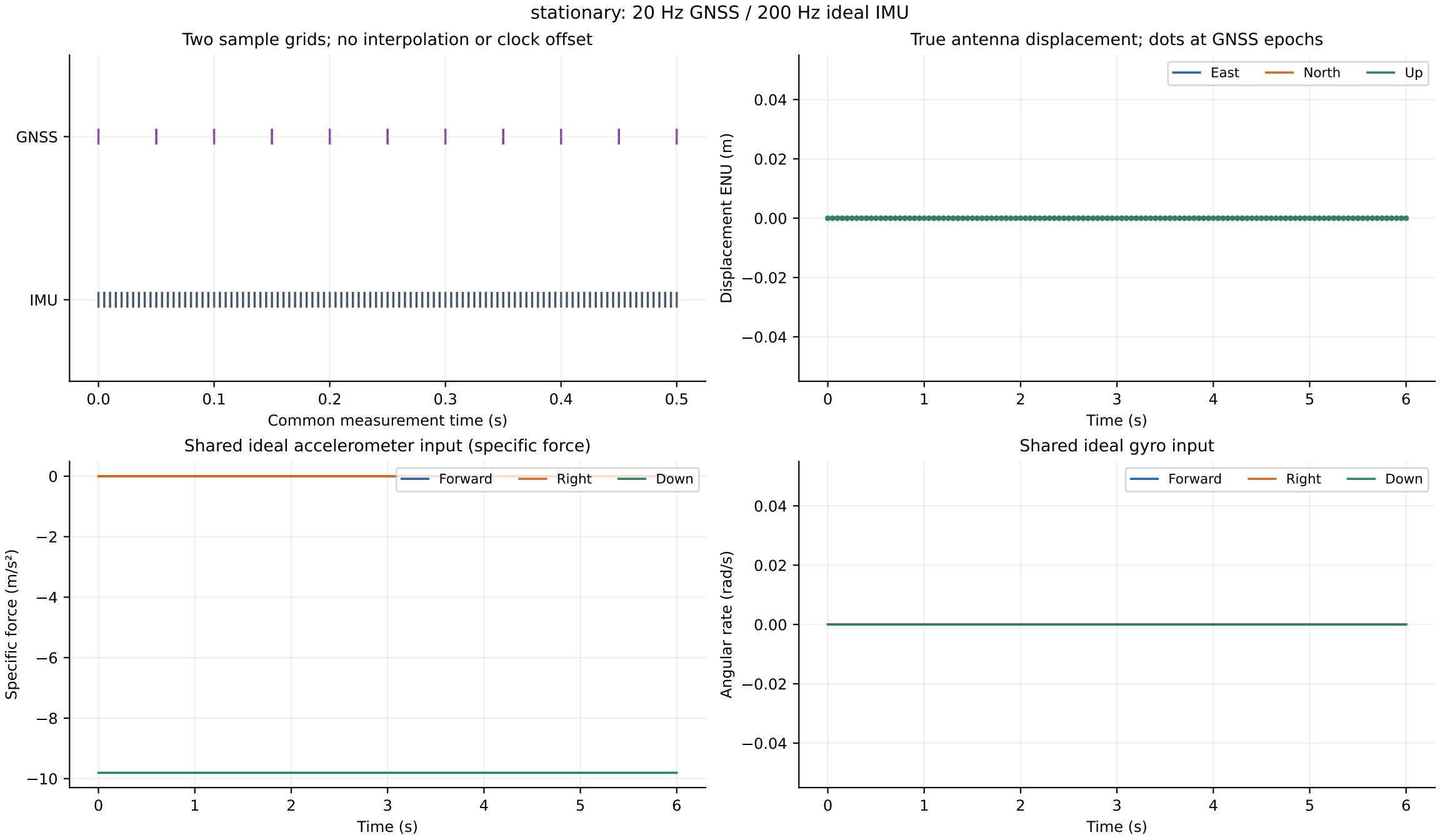
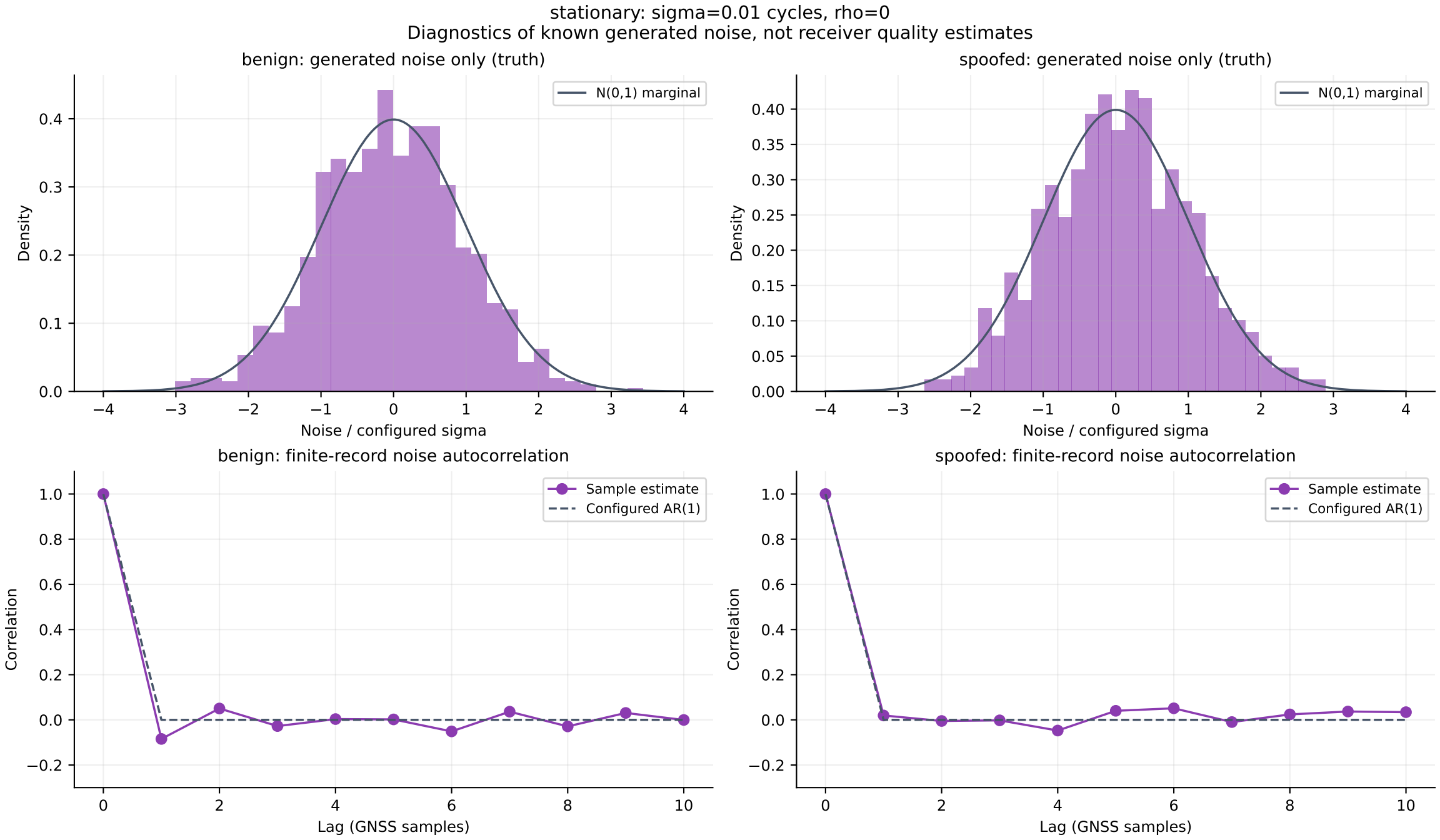
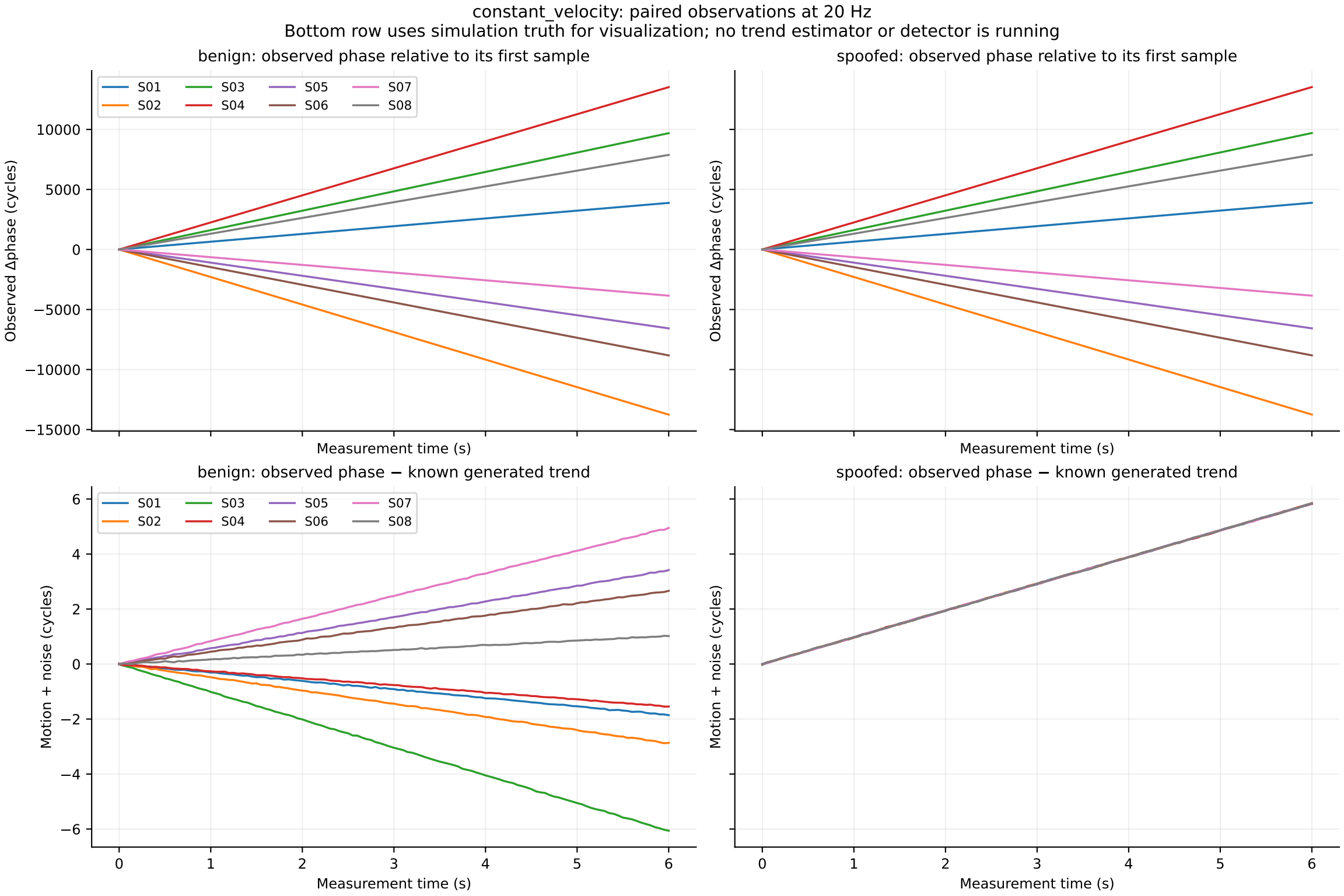

stationary: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
15000
10000
5000
0
−5000
−10000
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0
1
2
3
4
5
6

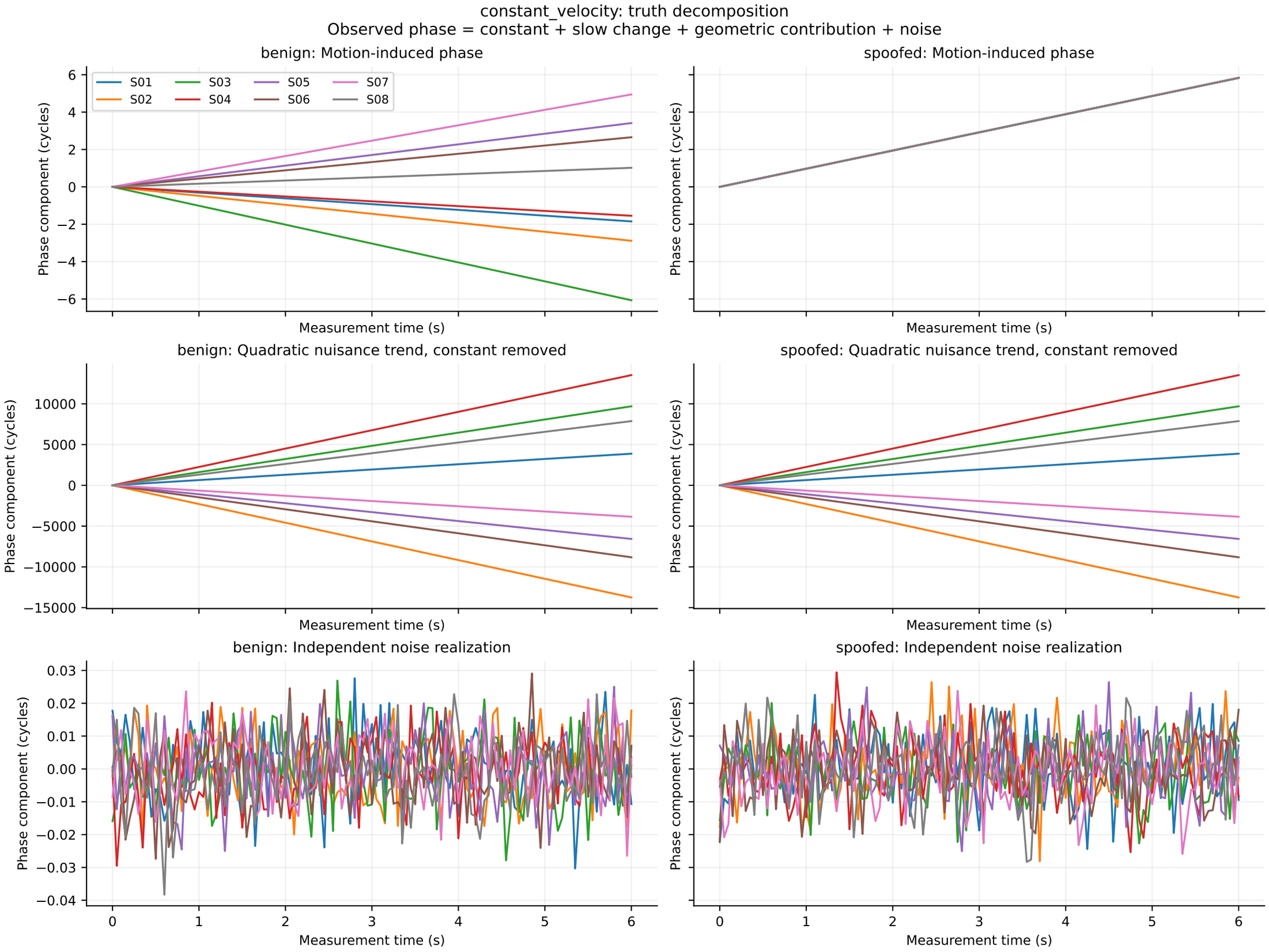

stationary: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
S01
S02
S03
S04
S05
S06
S07
S08
Phase component (cycles)
Measurement time (s)
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

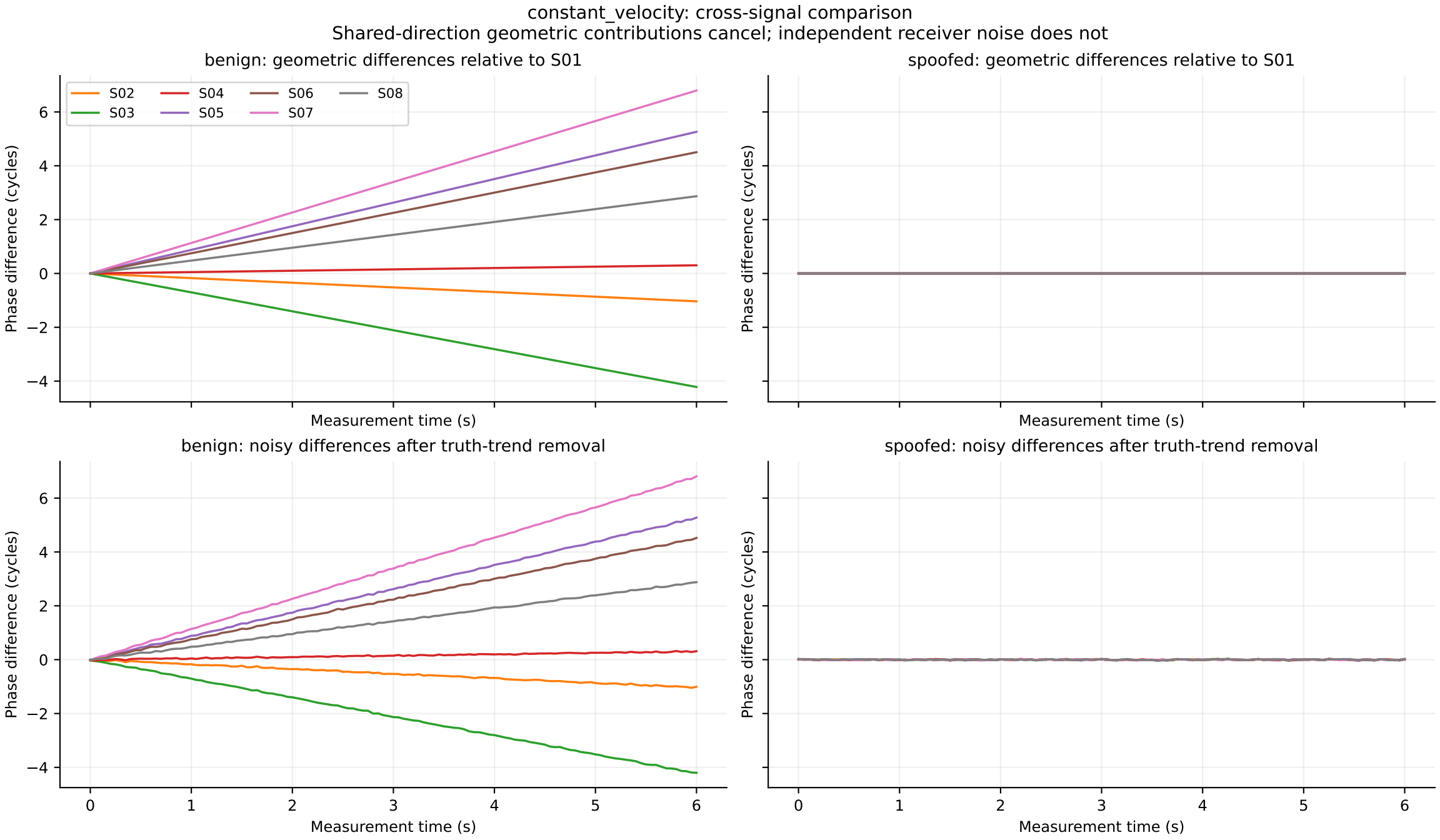

stationary: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
S02
S03
S04
S05
S06
S07
S08
Phase difference (cycles)
Measurement time (s)
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6

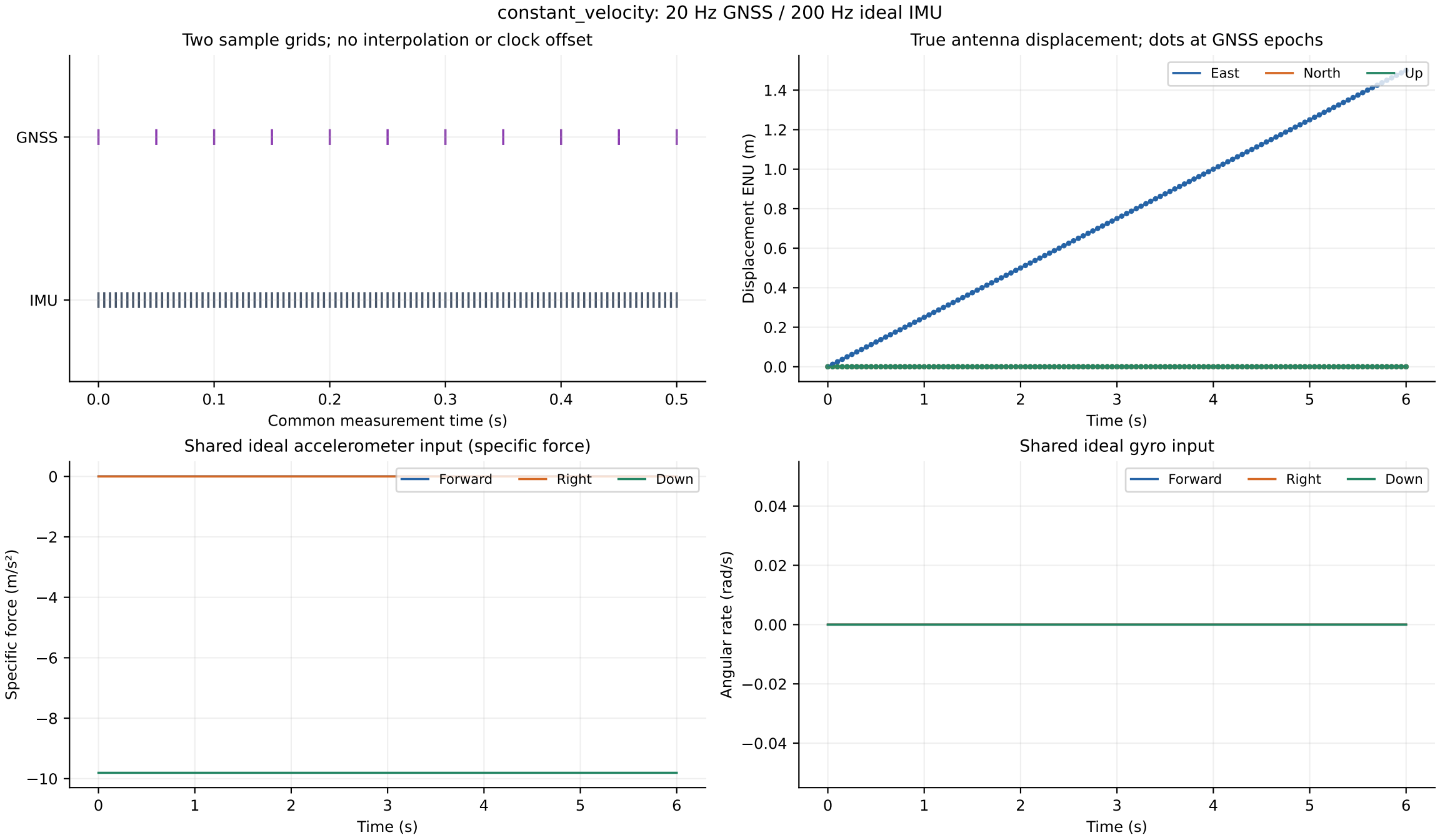

stationary: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
0
−2
−4
−6
−8
−10
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
0.04
0.02
0.00
−0.02
−0.04
Time (s)

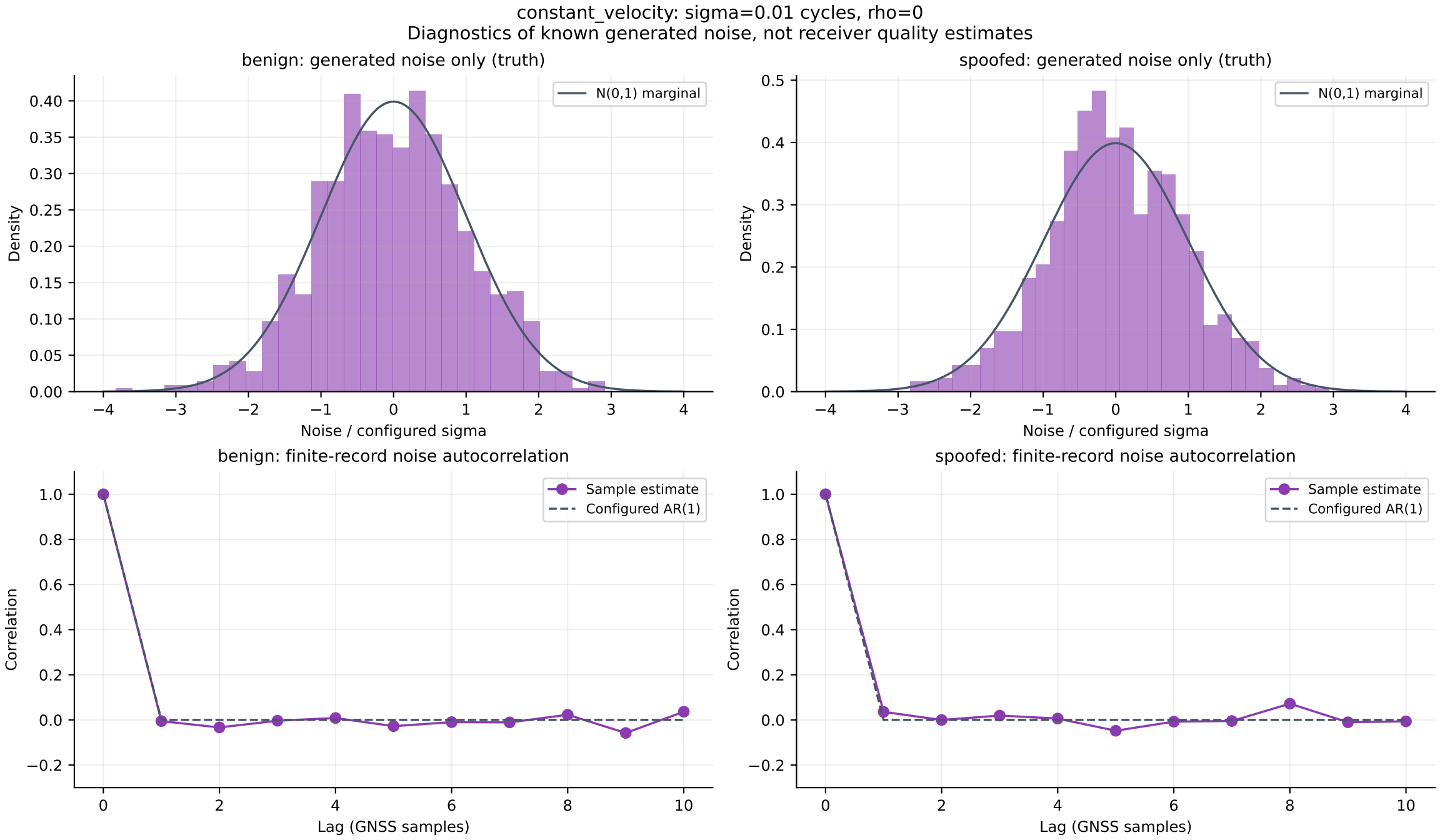

stationary: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

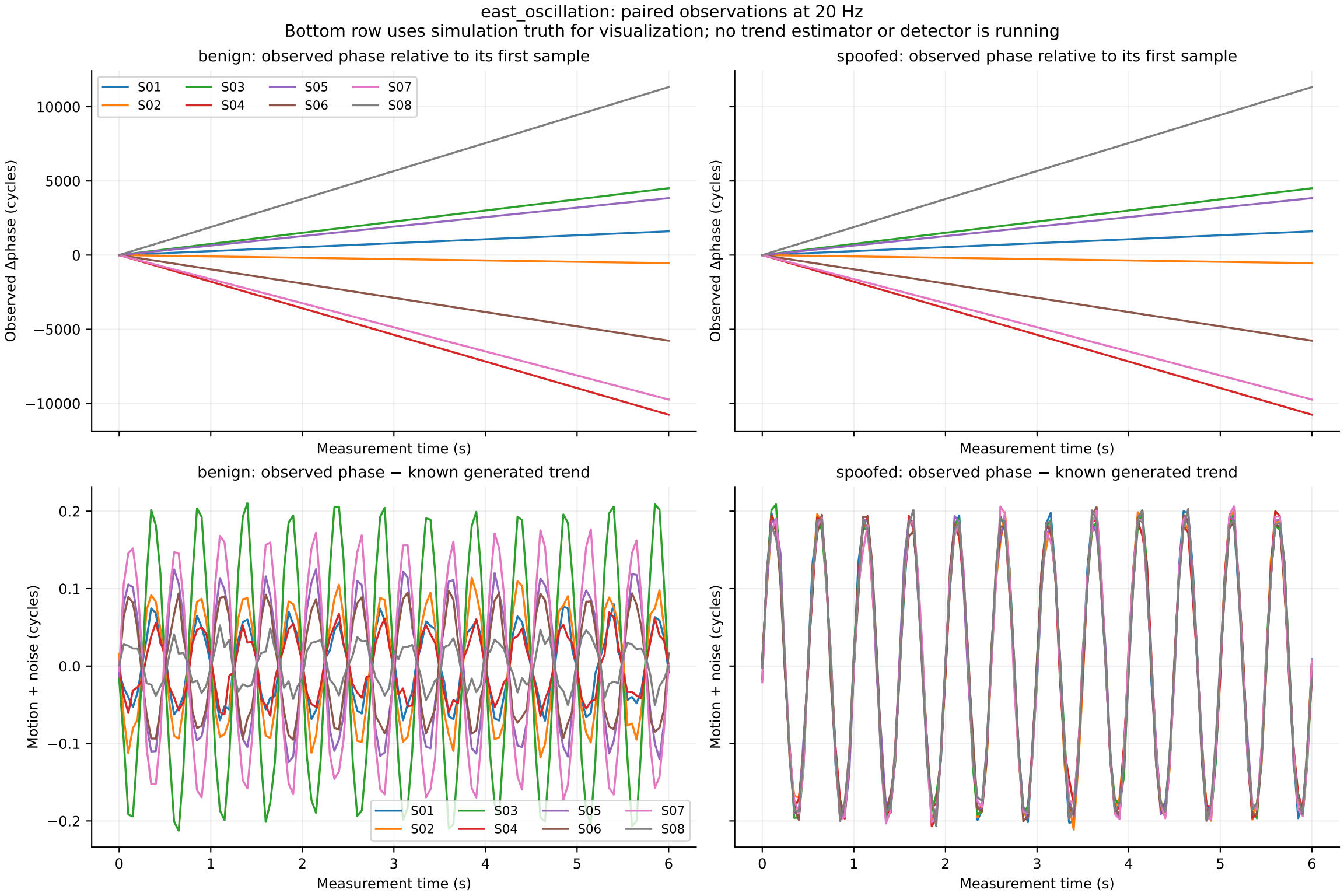

constant_velocity: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
10000
5000
0
−5000
−10000
−15000
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
6
4
2
0
−2
−4
−6
0
1
2
3
4
5
6
Measurement time (s)

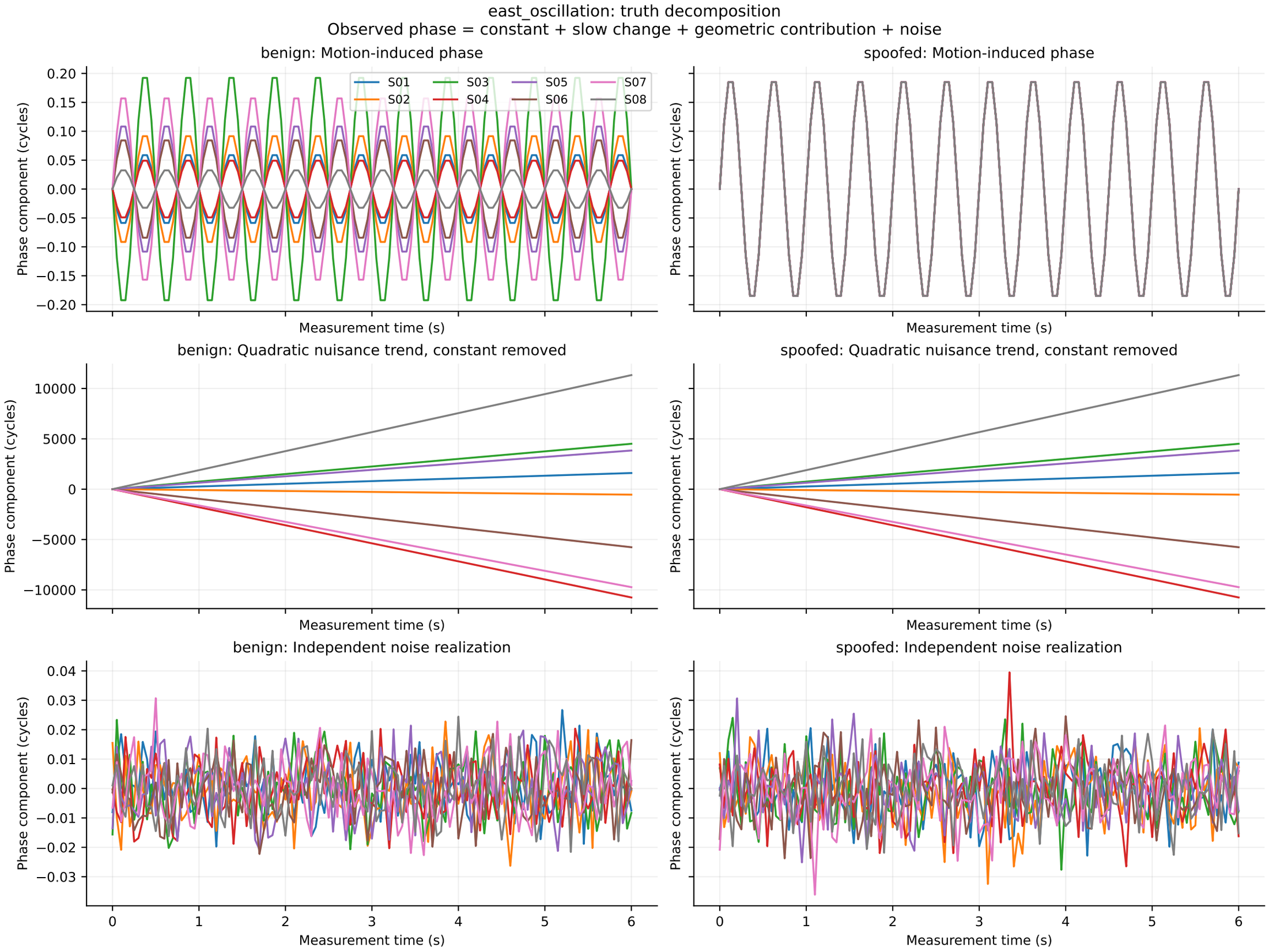

constant_velocity: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
S01
S02
S03
S04
S05
S06
S07
S08
Phase component (cycles)
Measurement time (s)
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

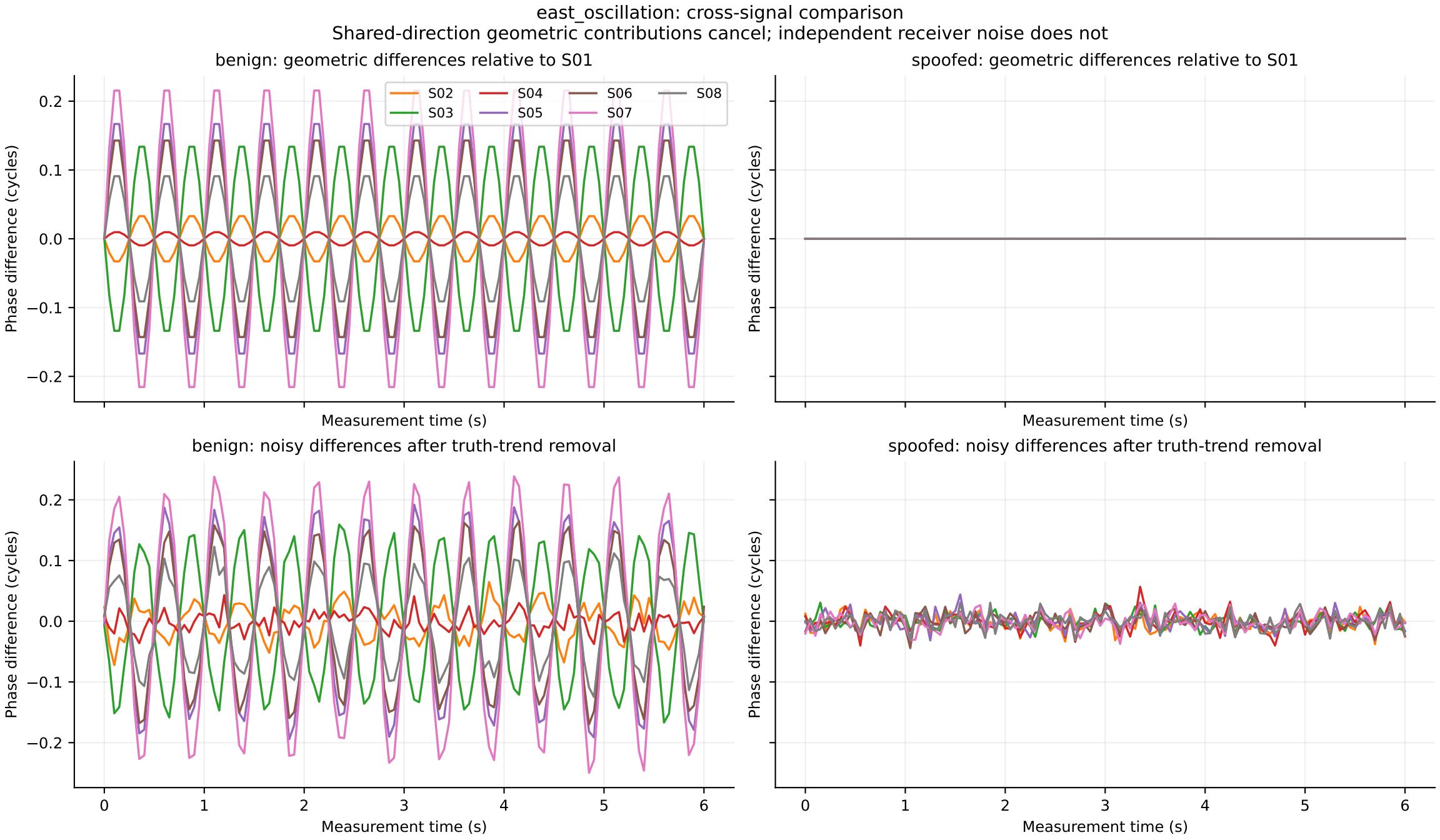

constant_velocity: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
S02
S03
S04
S05
S06
S07
S08
Phase difference (cycles)
Measurement time (s)
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal

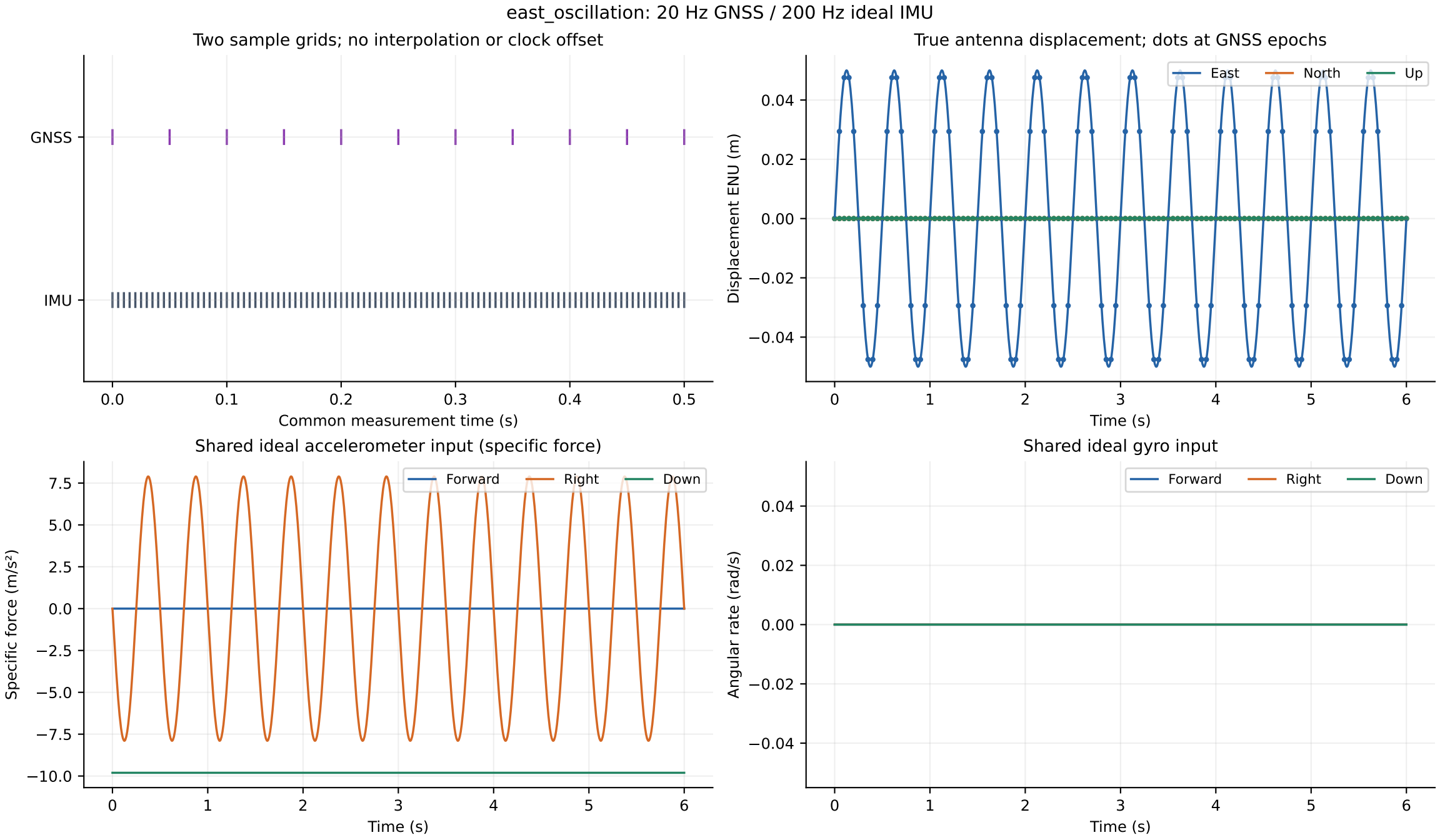

constant_velocity: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
0
−2
−4
−6
−8
−10
0
1
2
3
4
5
6
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6
Time (s)

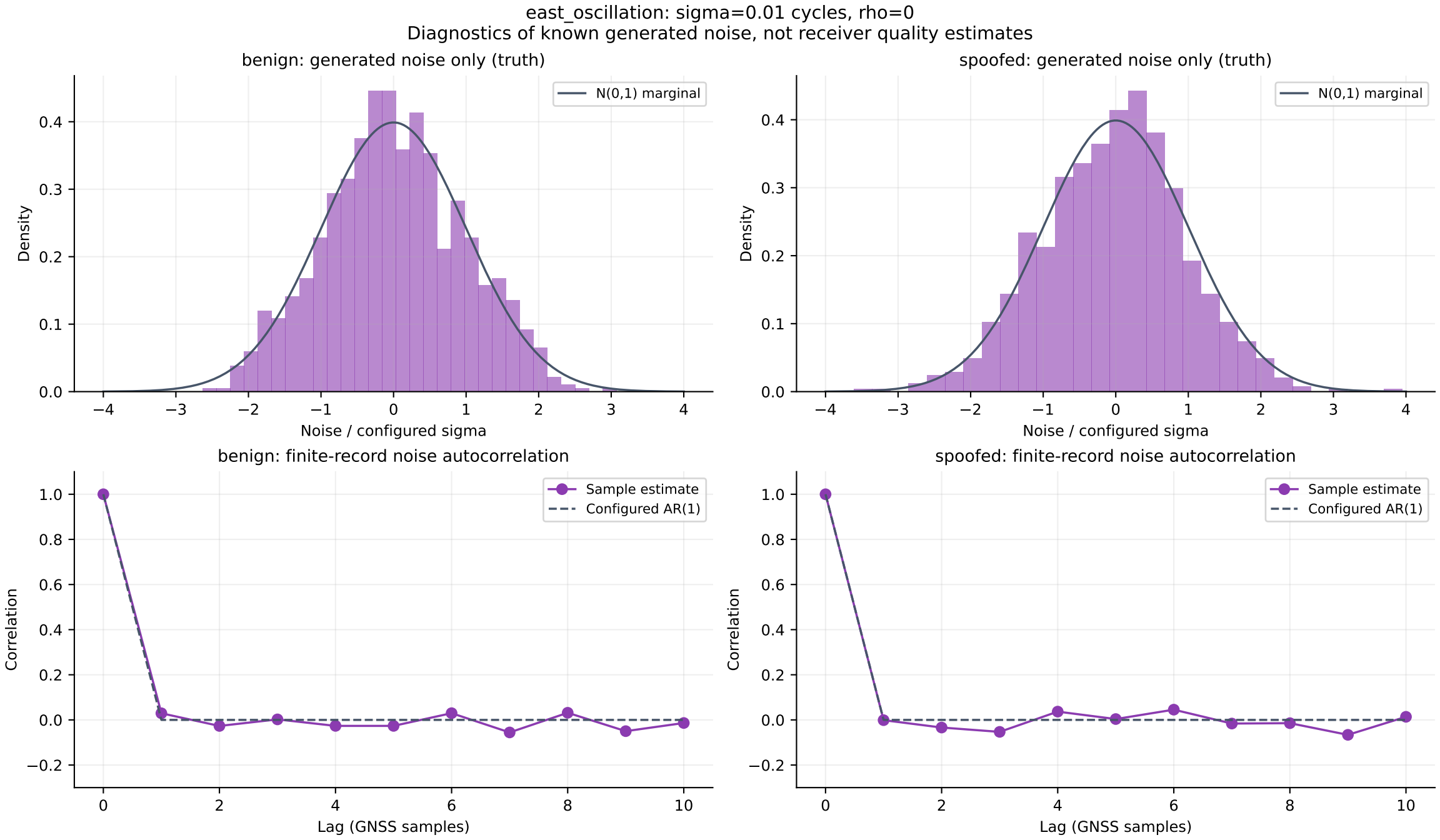

constant_velocity: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

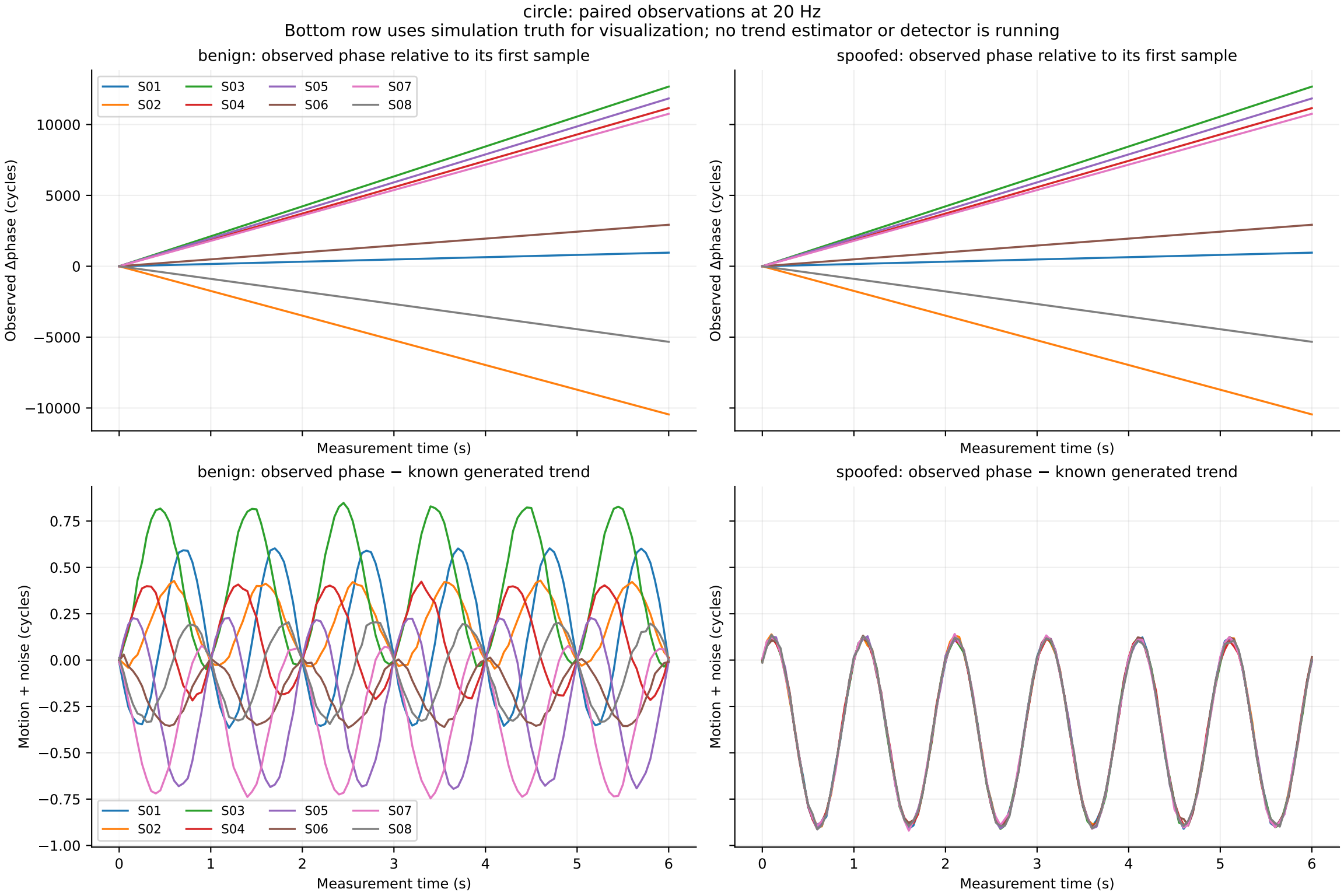

east_oscillation: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
Observed Δphase (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)

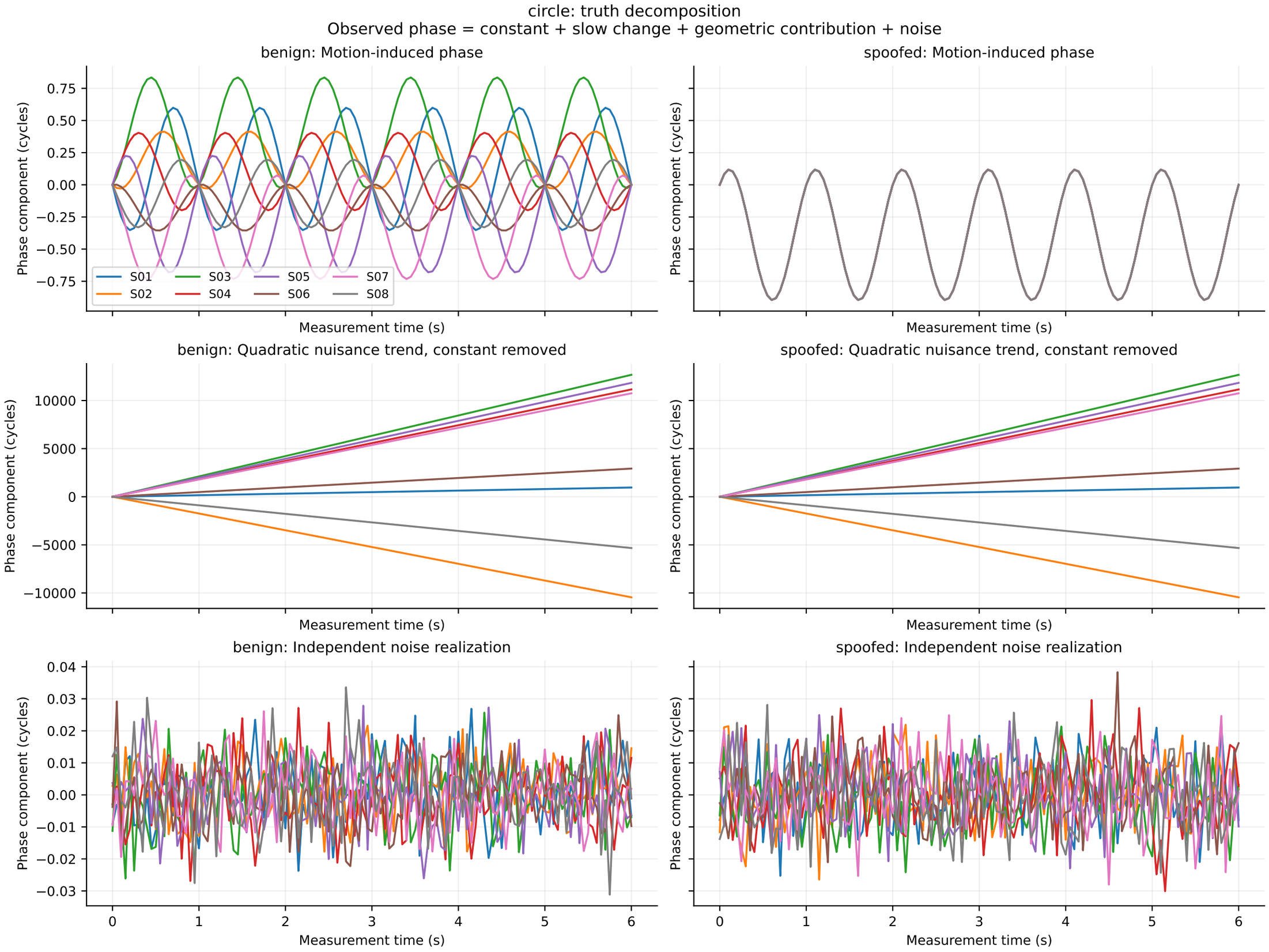

east_oscillation: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
S01
S02
S03
S04
S05
S06
S07
S08
Phase component (cycles)
Measurement time (s)
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

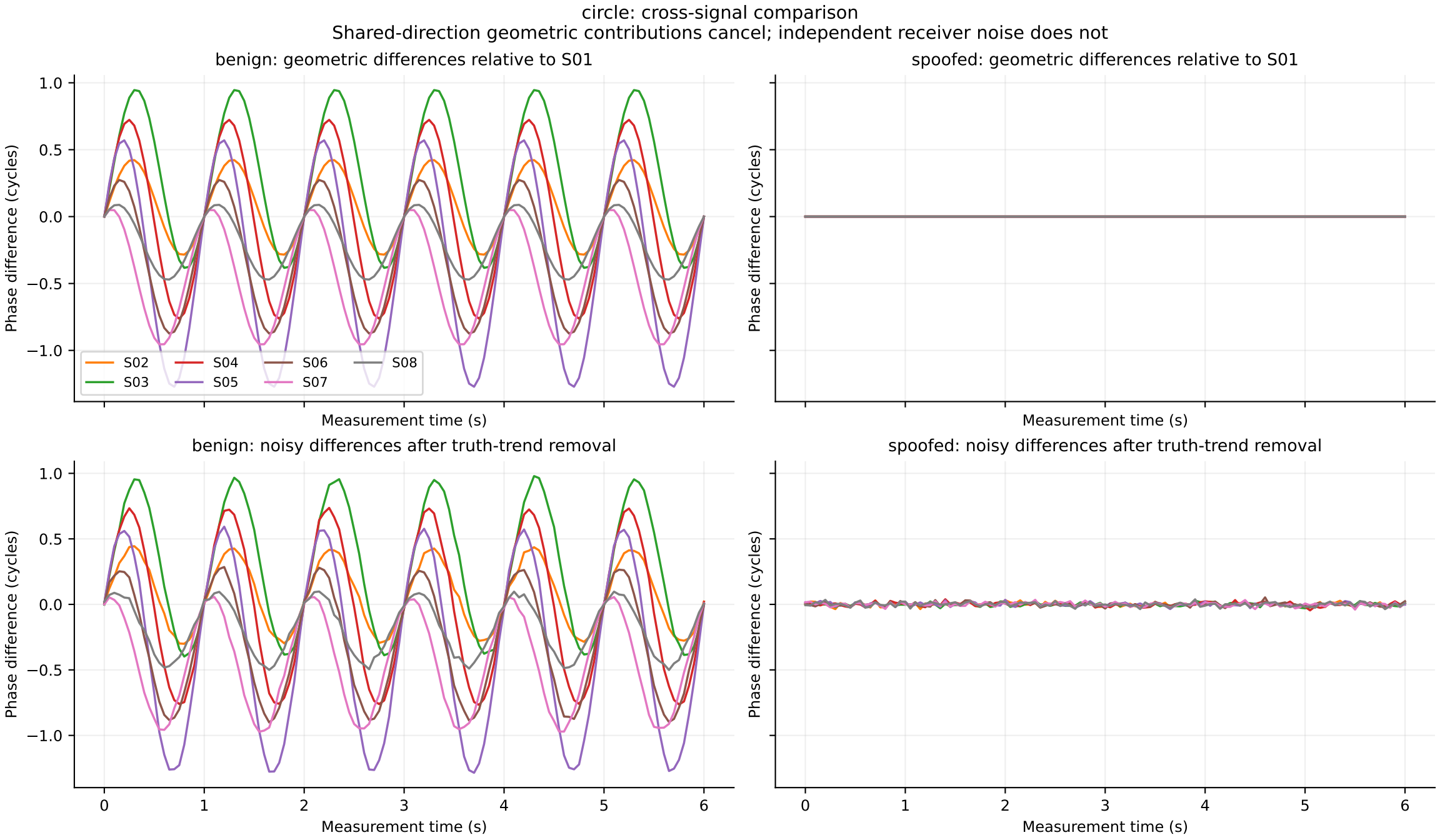

east_oscillation: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
S02
S03
S04
S05
S06
S07
S08
Phase difference (cycles)
Measurement time (s)
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal

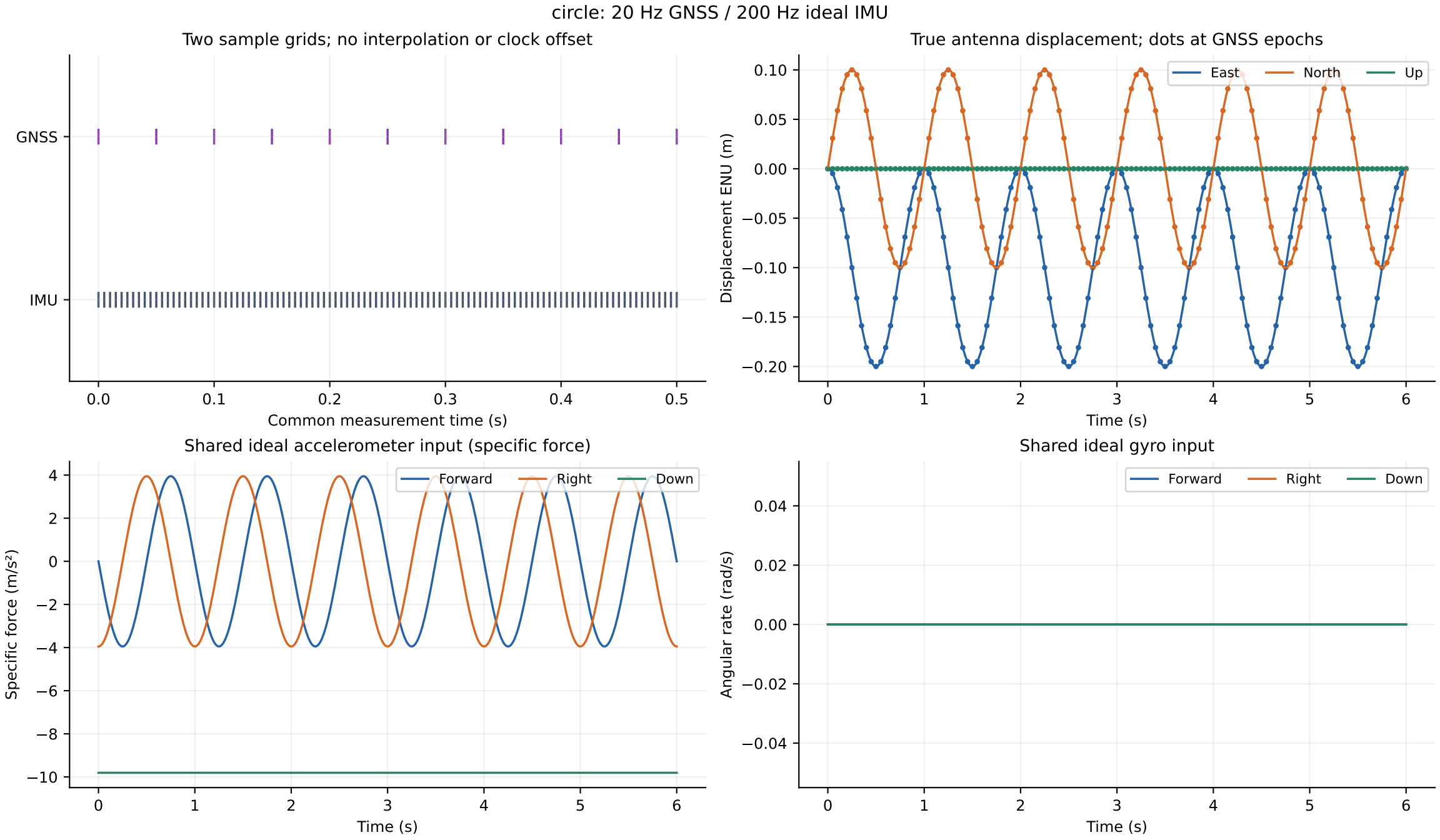

east_oscillation: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
0.04
0.02
0.00
−0.02
−0.04
Time (s)

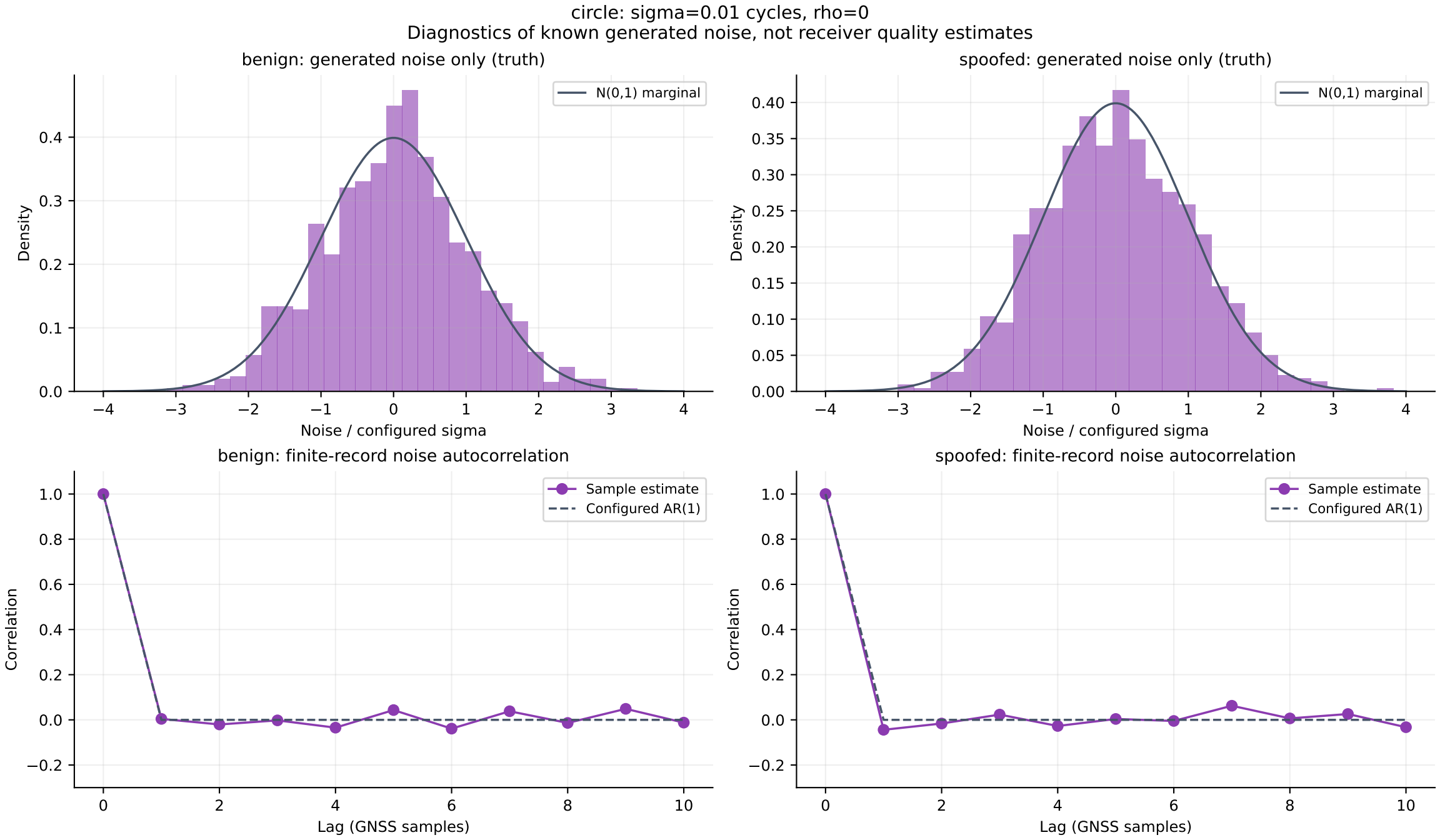

east_oscillation: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

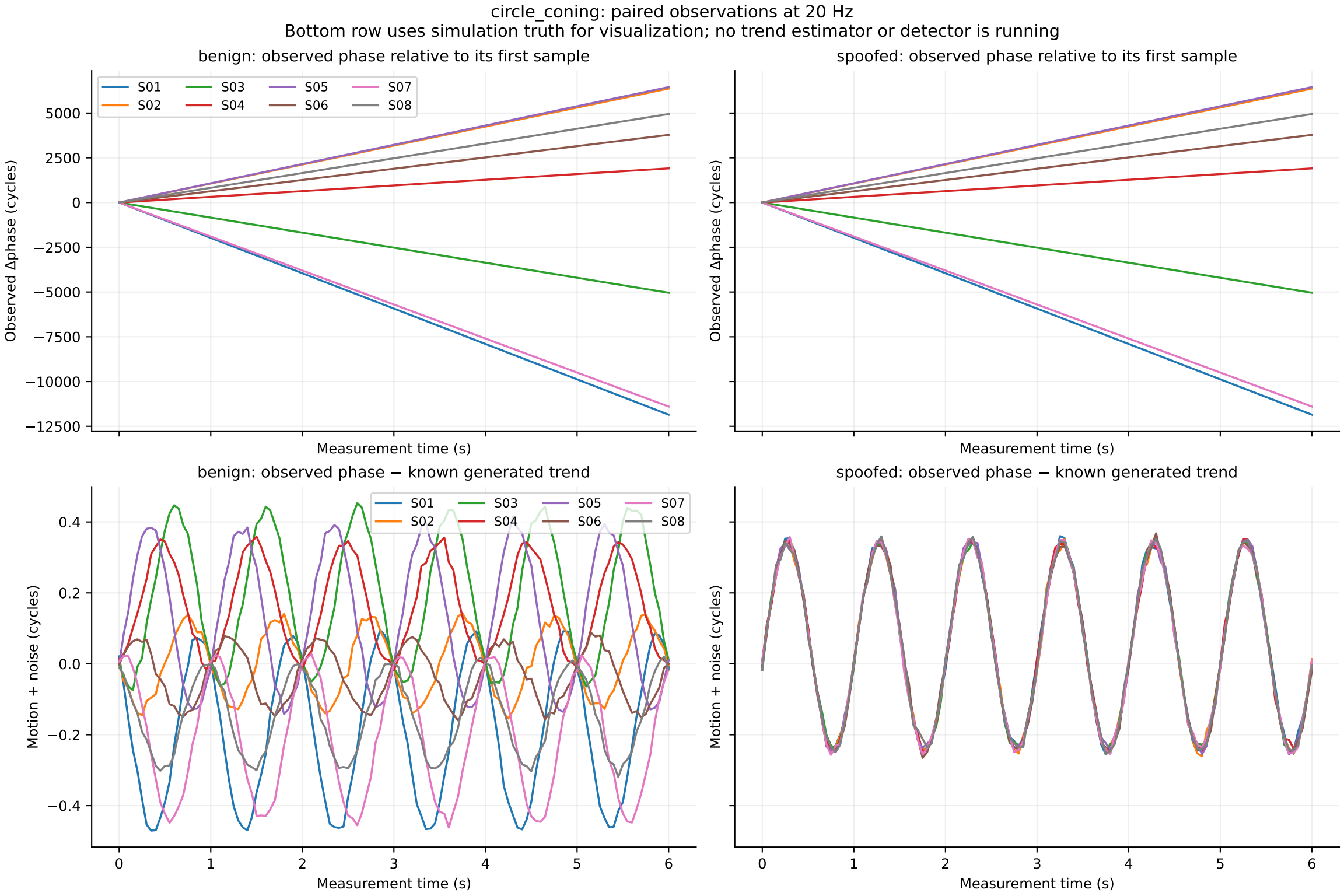

circle: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
10000
5000
0
−5000
−10000
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
6

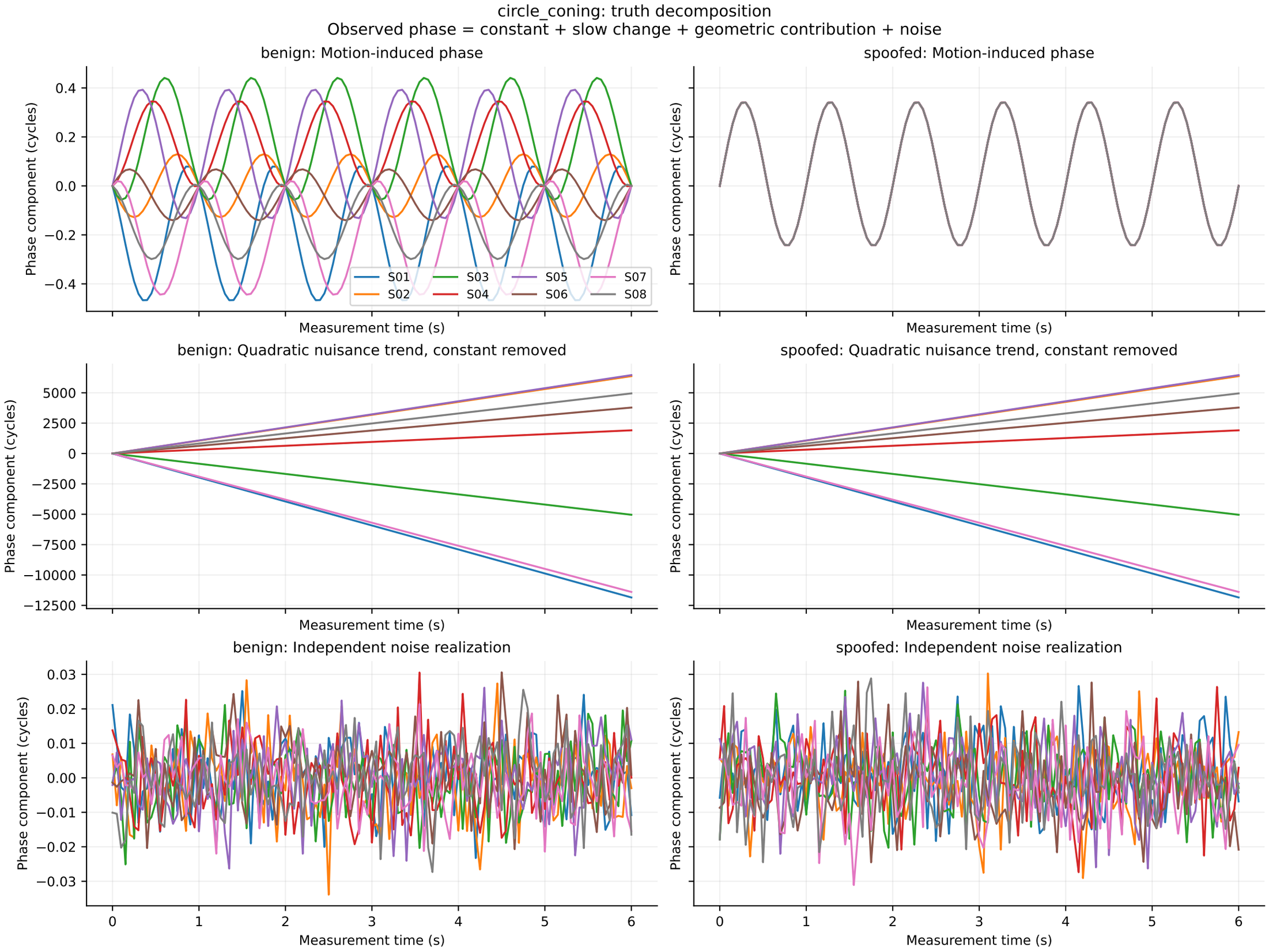

circle: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
Phase component (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

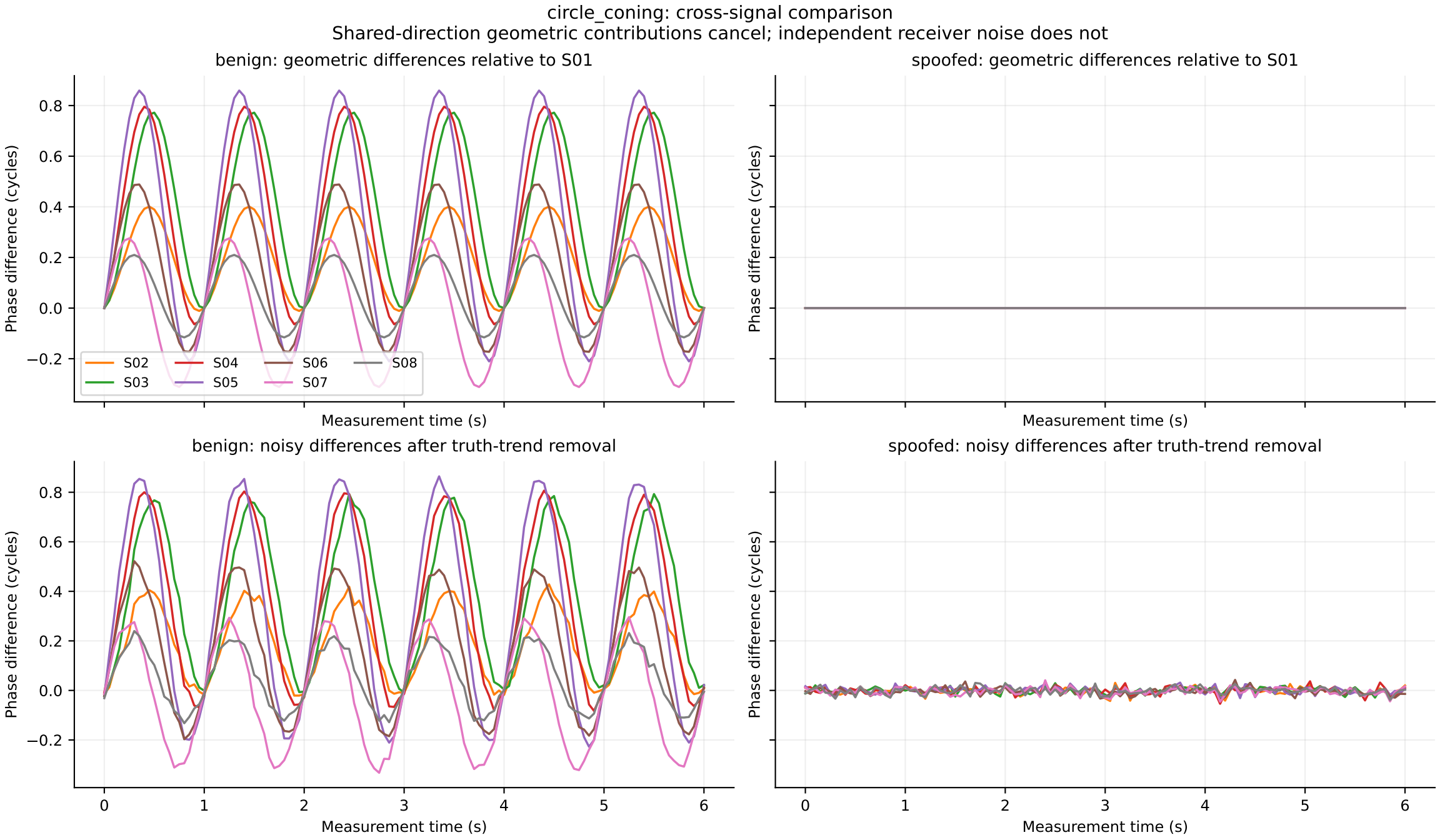

circle: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
Phase difference (cycles)
Measurement time (s)
S02
S03
S04
S05
S06
S07
S08

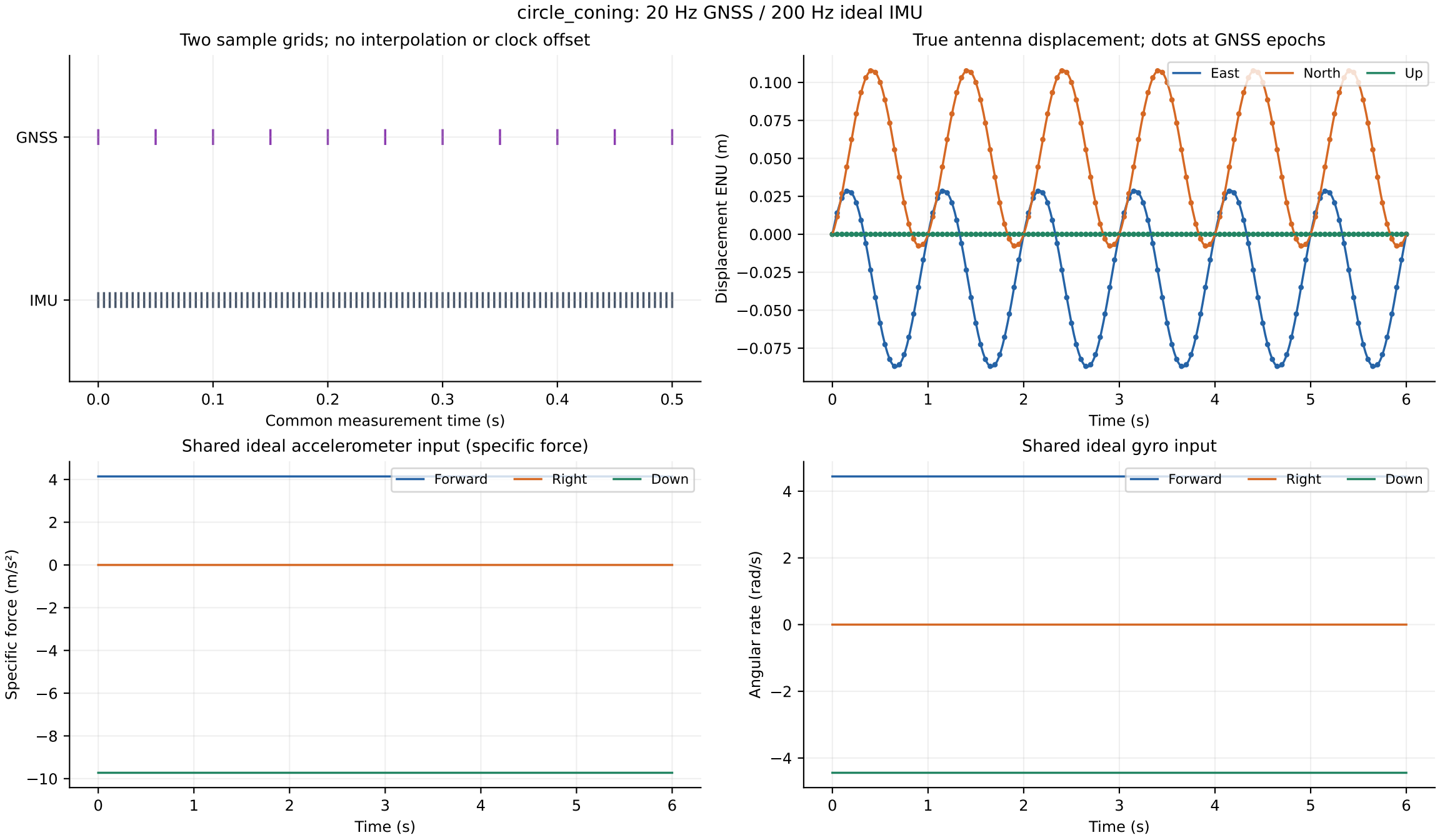

circle: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
4
2
0
−2
−4
−6
−8
−10
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
0.04
0.02
0.00
−0.02
−0.04
Time (s)

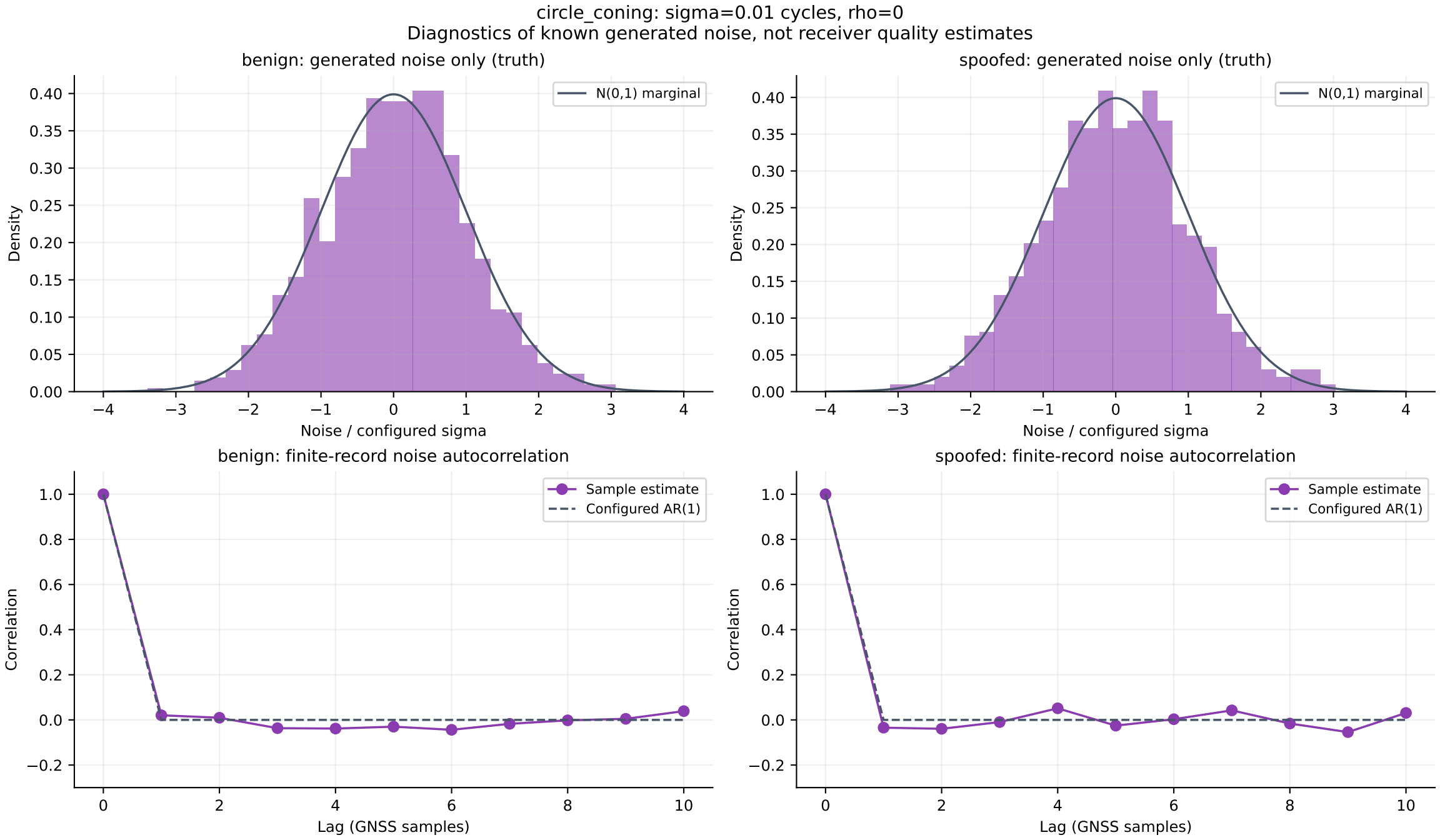

circle: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

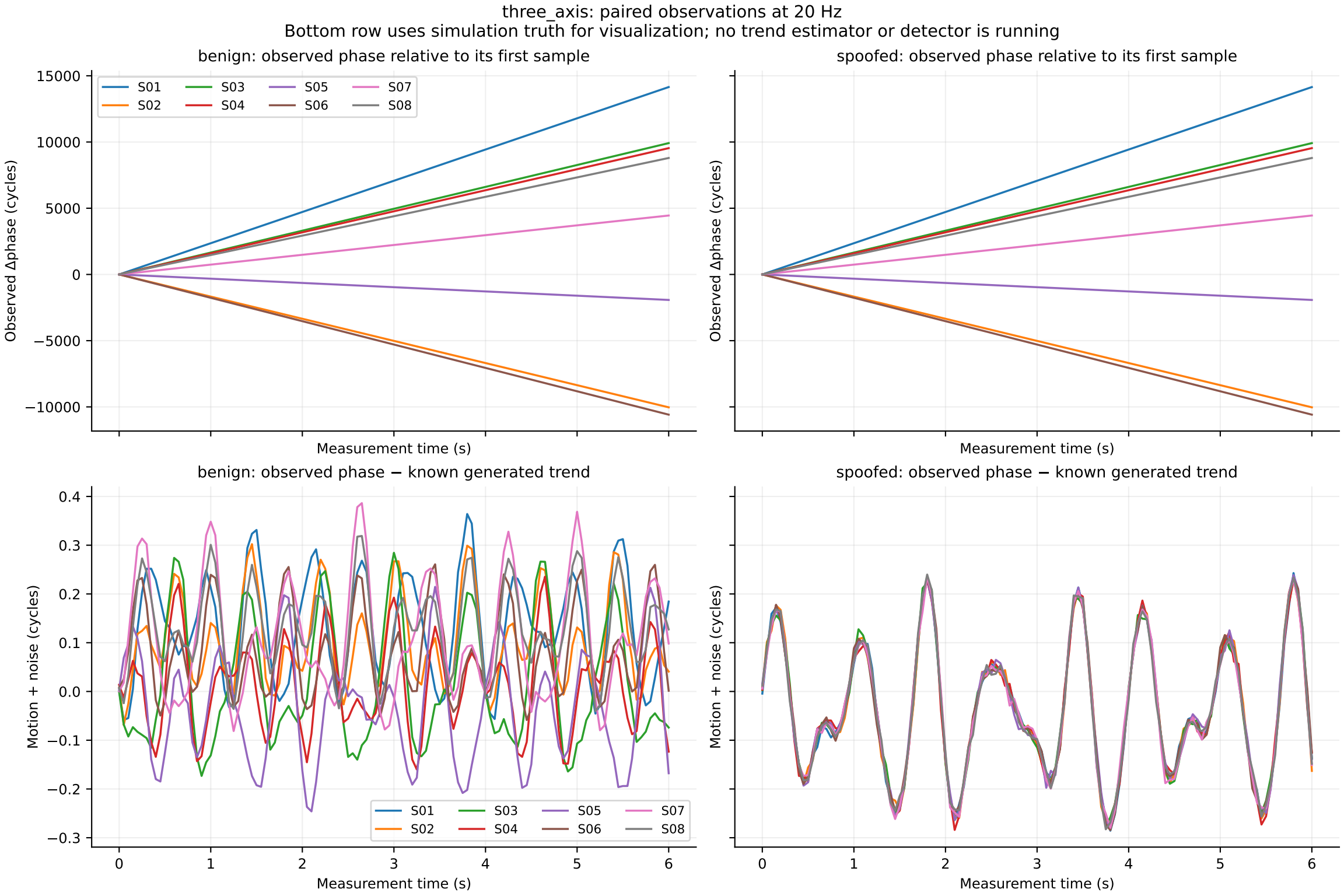

circle_coning: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
5000
2500
0
−2500
−5000
−7500
−10000
−12500
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
0.4
0.2
0.0
−0.2
−0.4
0
1
2
3
4
5
6
Measurement time (s)

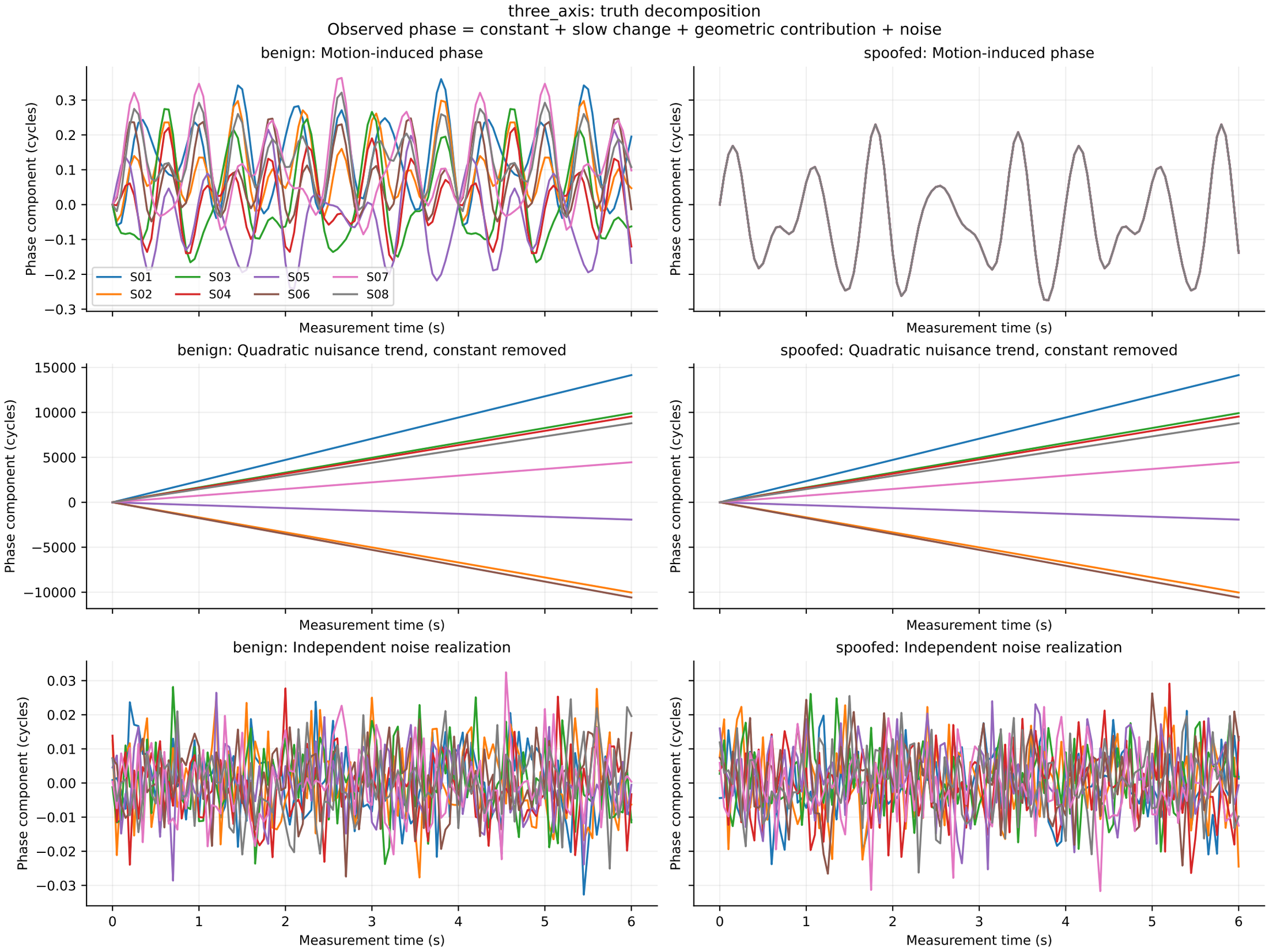

circle_coning: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
Phase component (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

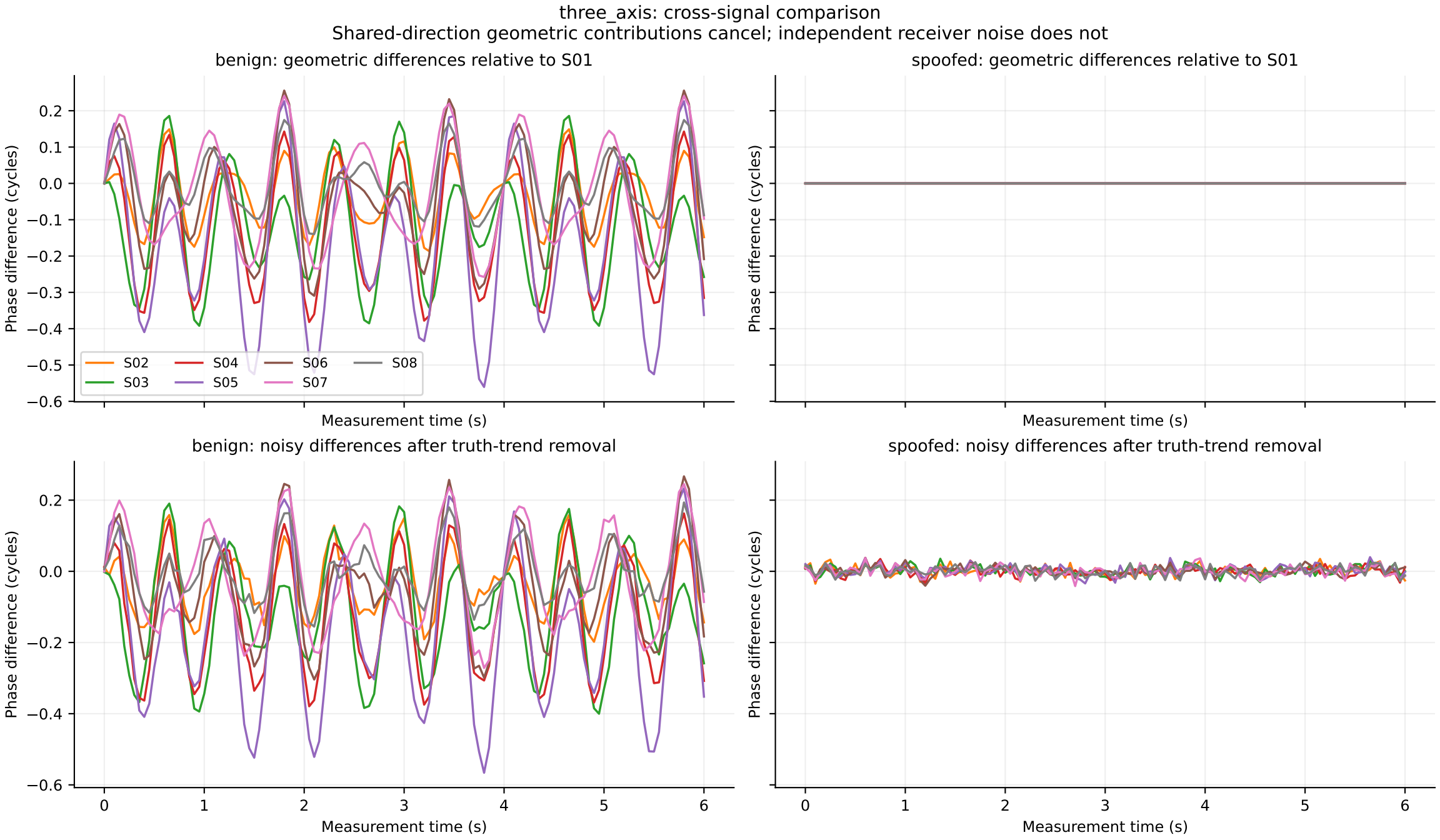

circle_coning: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
Phase difference (cycles)
Measurement time (s)
S02
S03
S04
S05
S06
S07
S08
0.8
0.6
0.4
0.2
0.0
−0.2
0
1
2
3
4
5
6

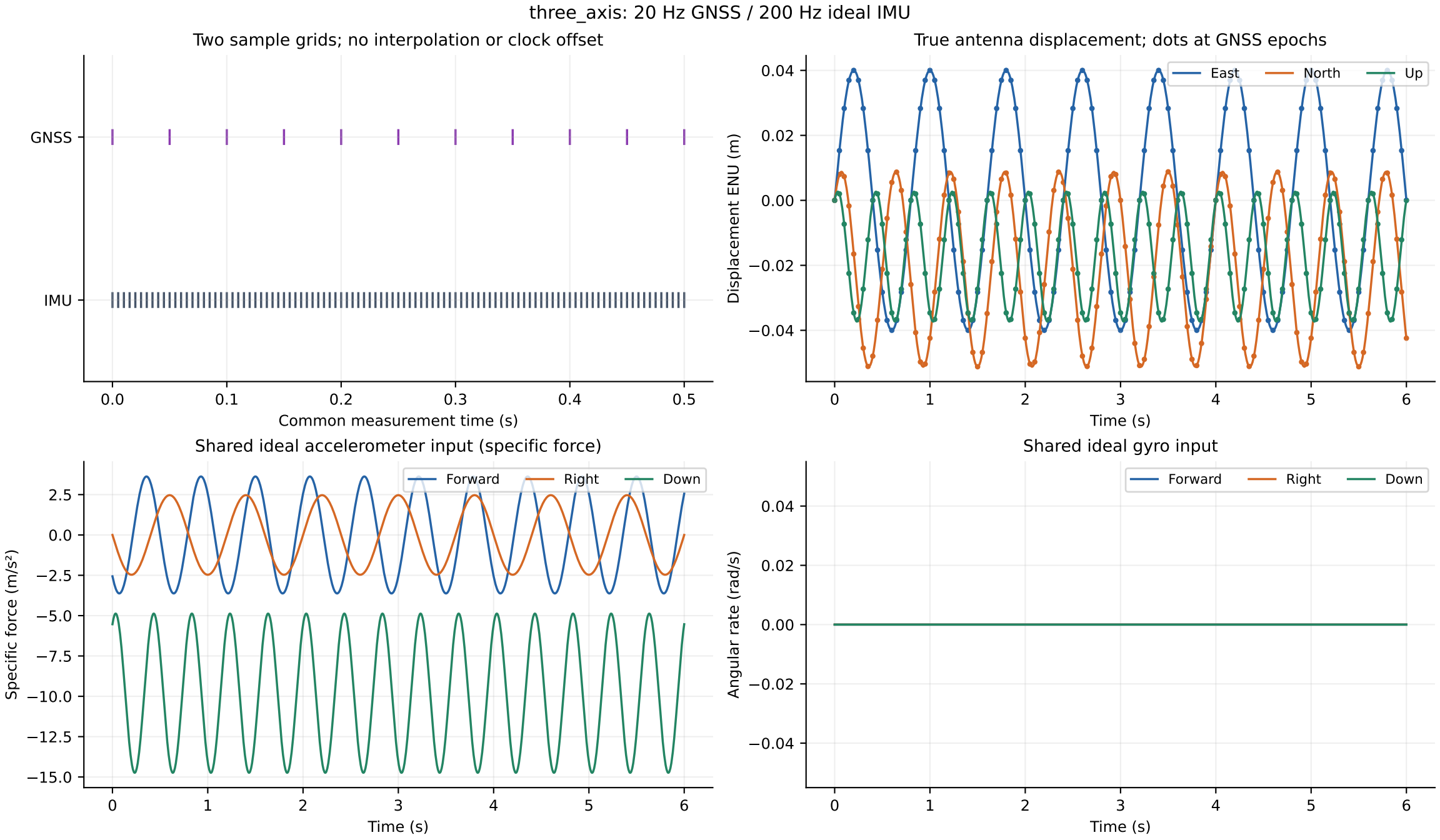

circle_coning: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
4
2
0
−2
−4
−6
−8
−10
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
4
2
0
−2
−4
Time (s)

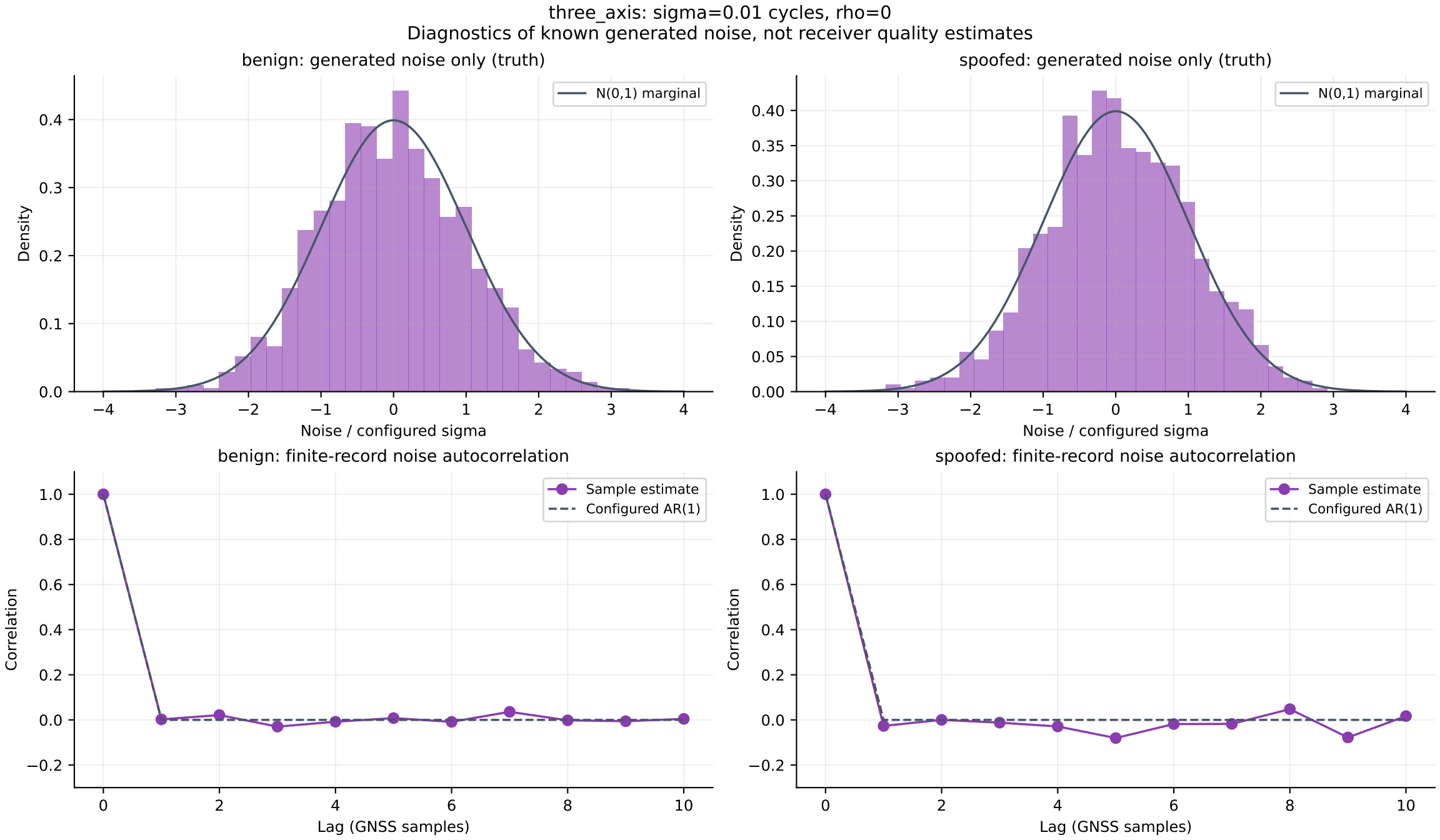

circle_coning: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

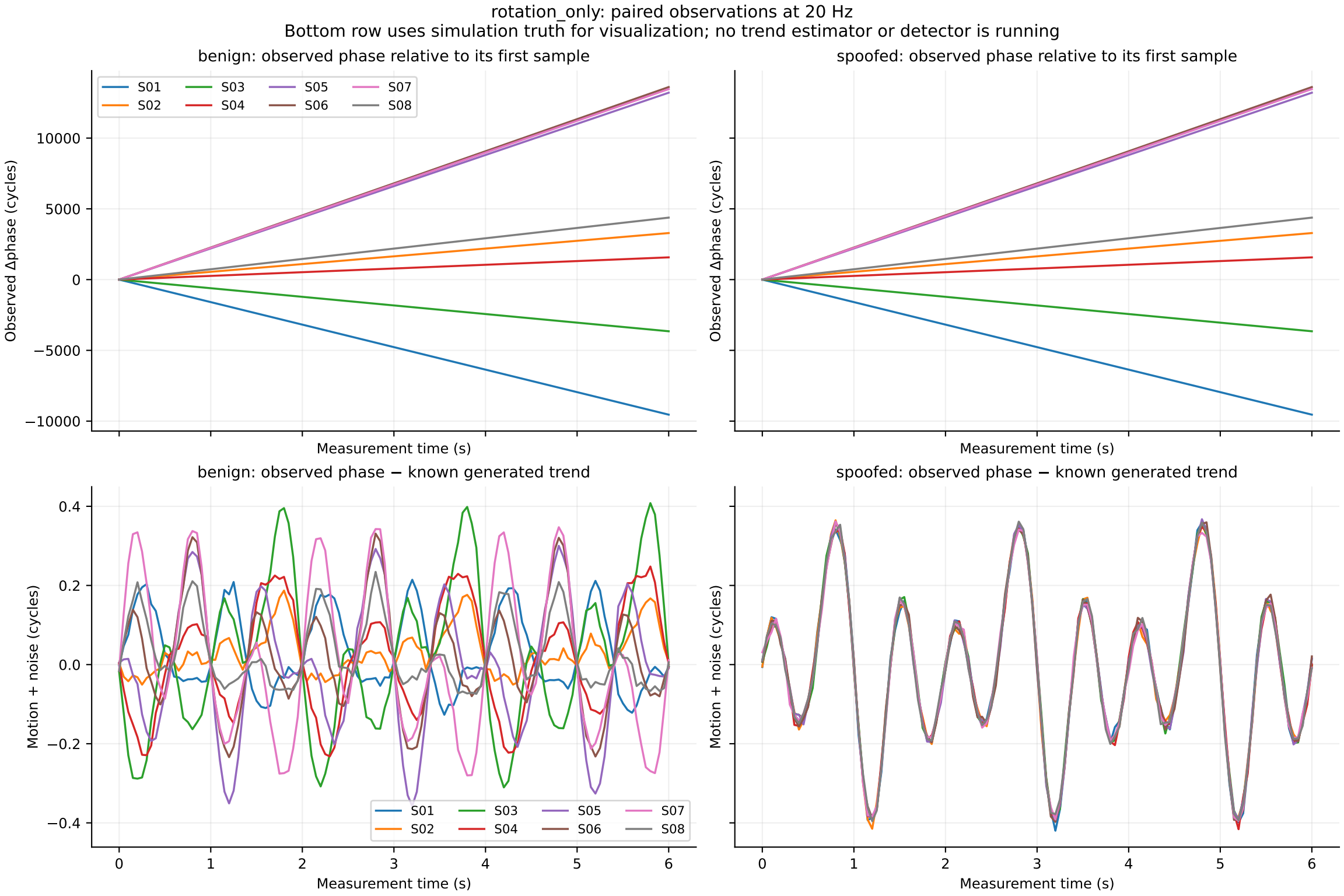

three_axis: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
Observed Δphase (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)

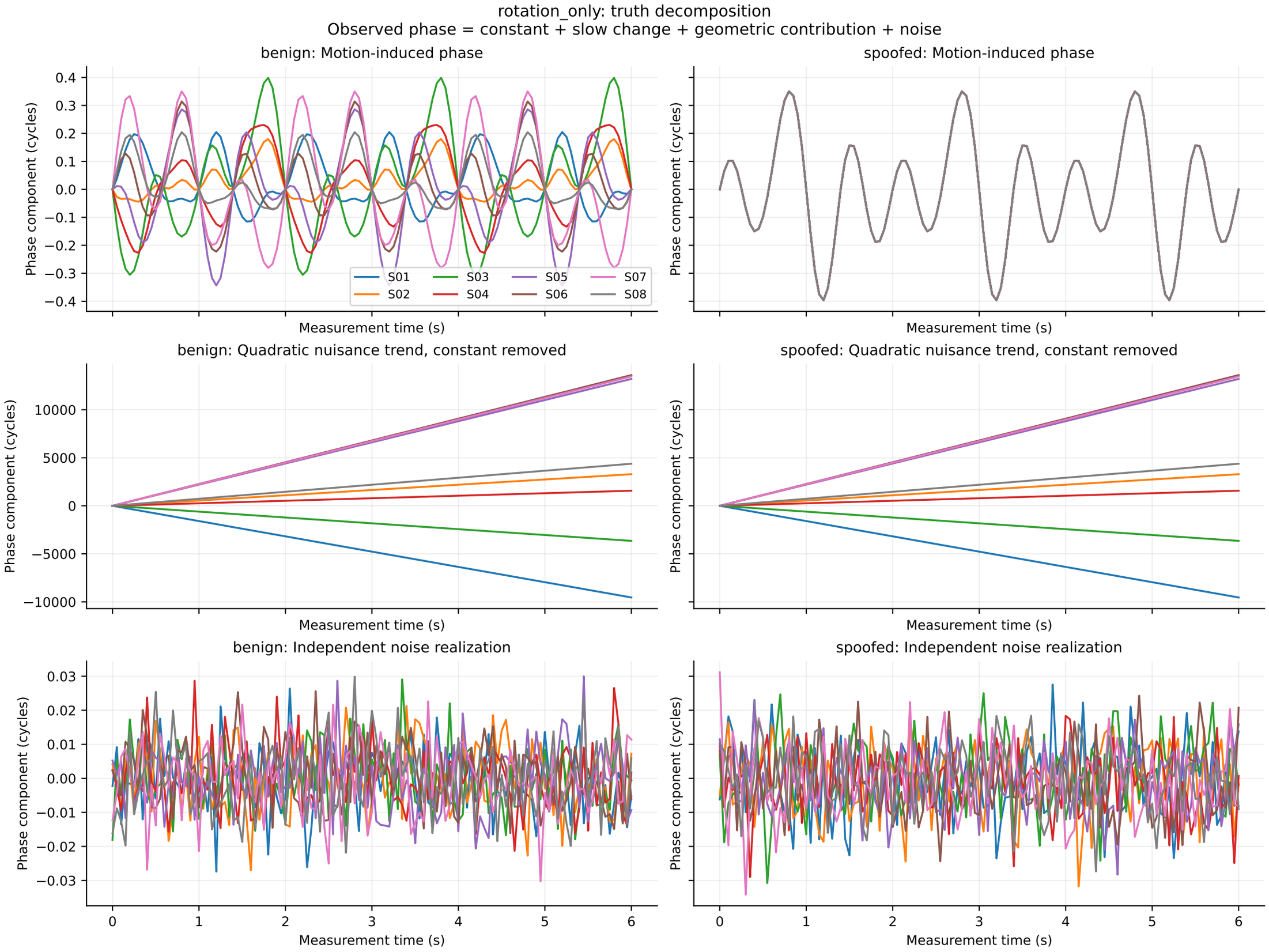

three_axis: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
Phase component (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization

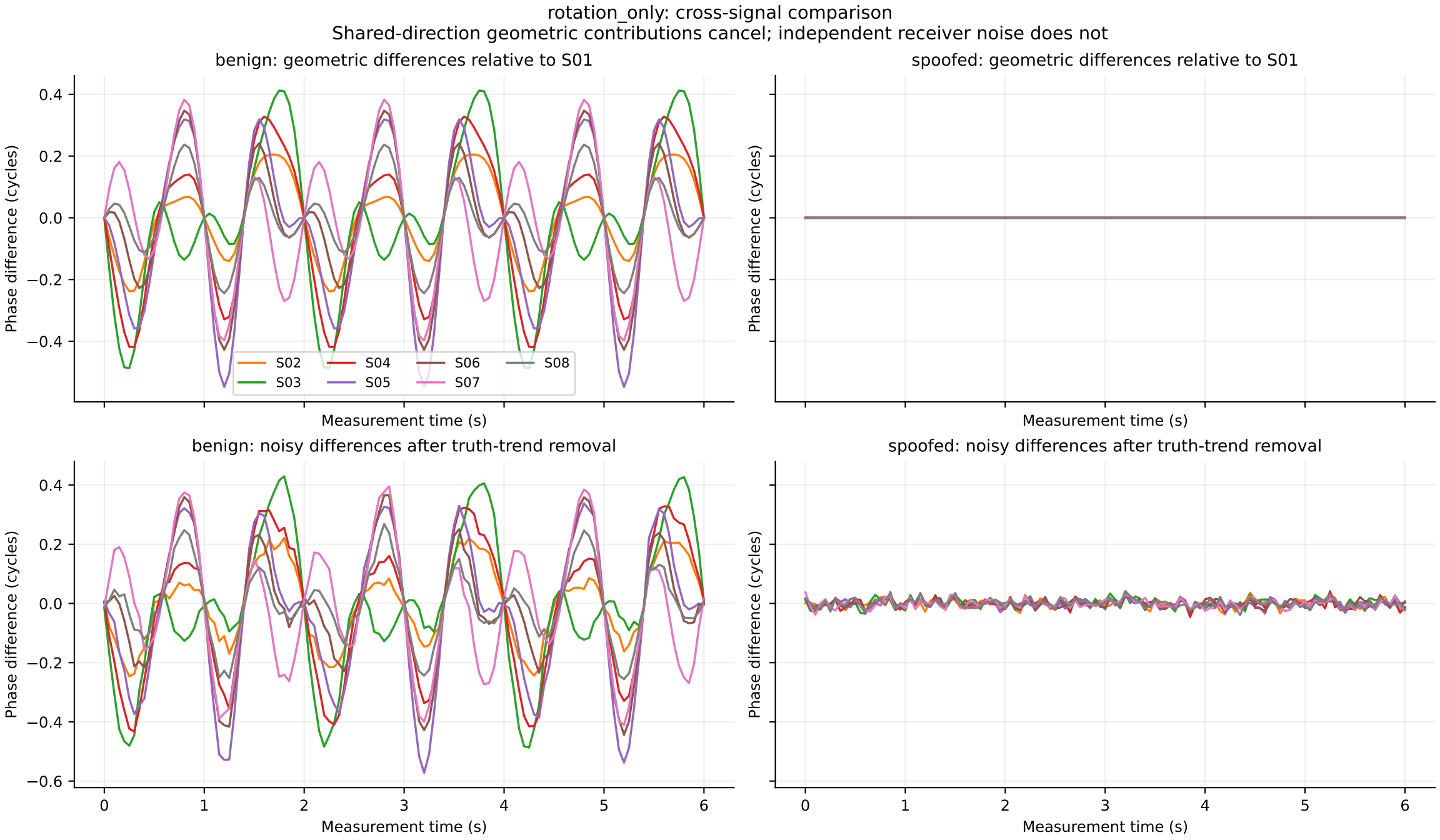

three_axis: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
Phase difference (cycles)
Measurement time (s)
S02
S03
S04
S05
S06
S07
S08

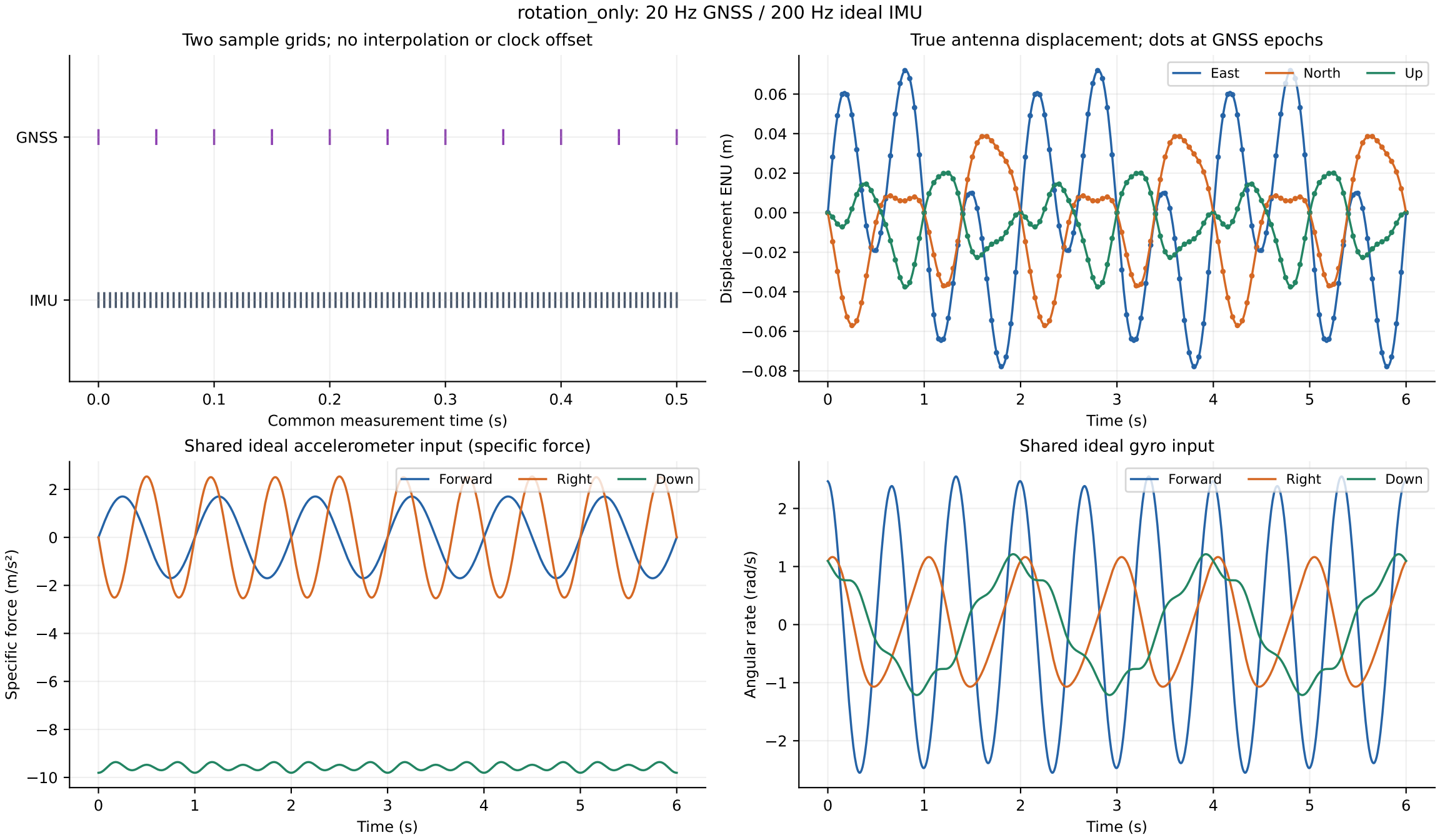

three_axis: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
0.04
0.02
0.00
−0.02
−0.04
Time (s)

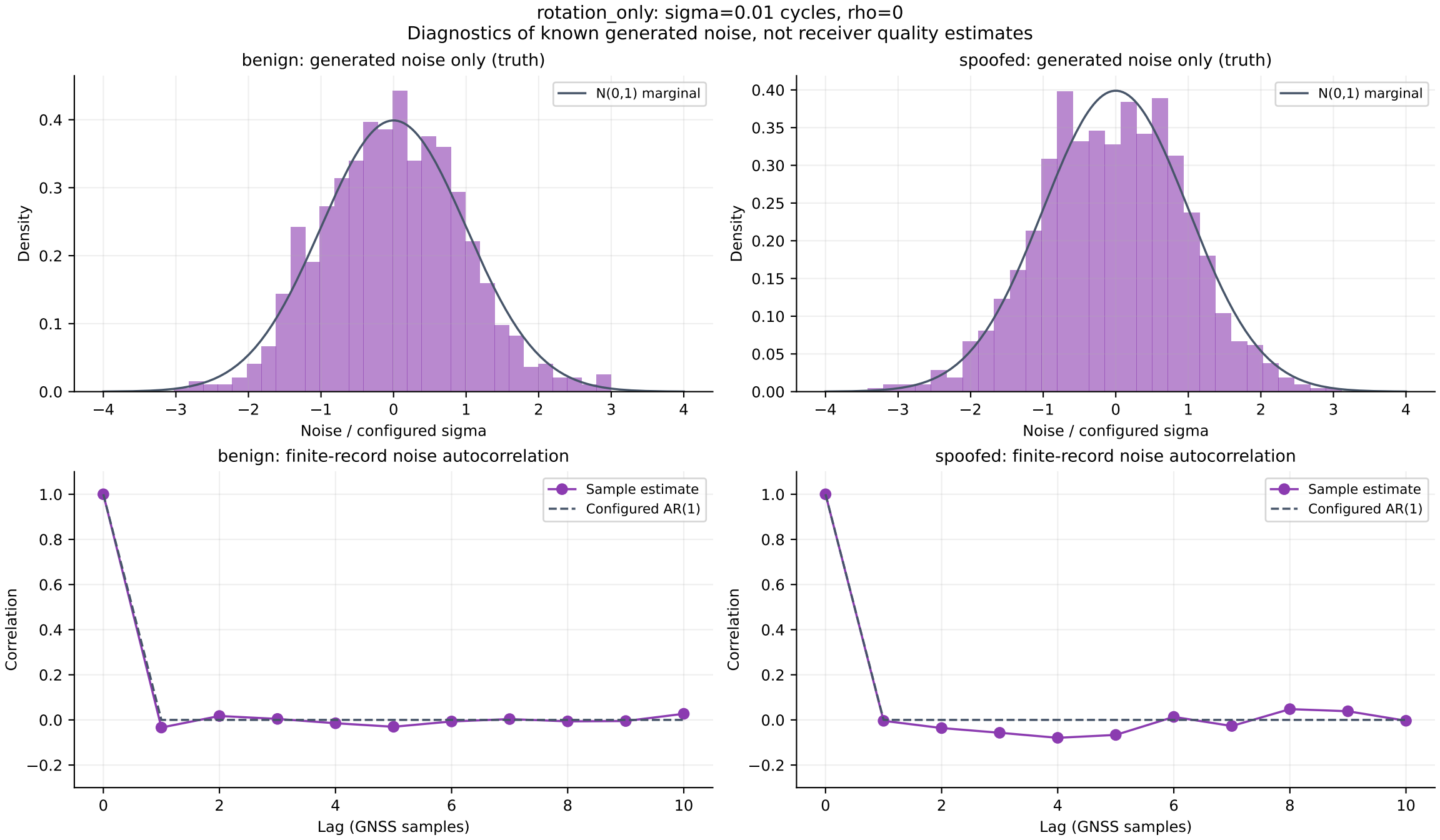

three_axis: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)

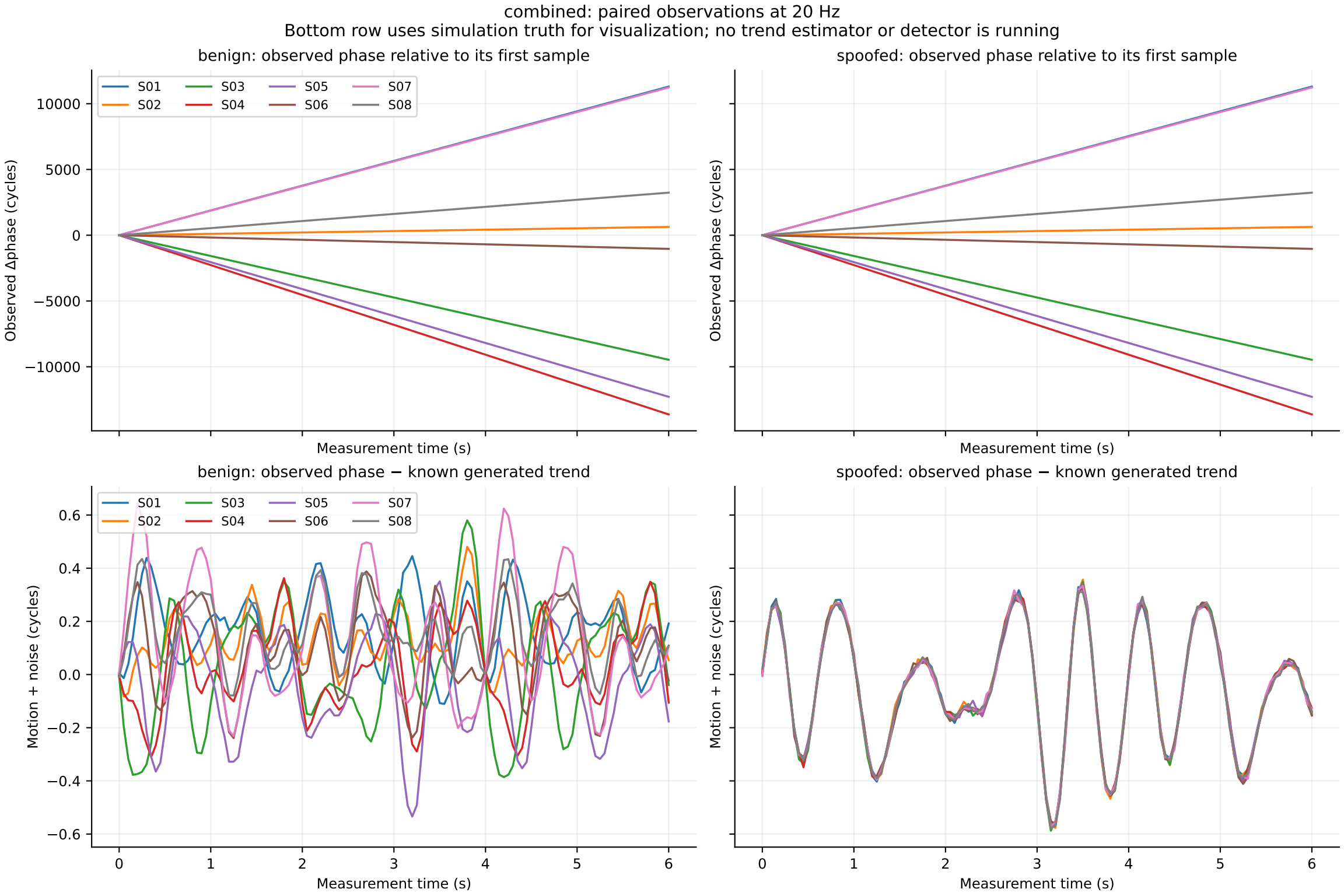

rotation_only: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
10000
5000
0
−5000
−10000
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
0.4
0.2
0.0
−0.2
−0.4
0
1
2
3
4
5
6

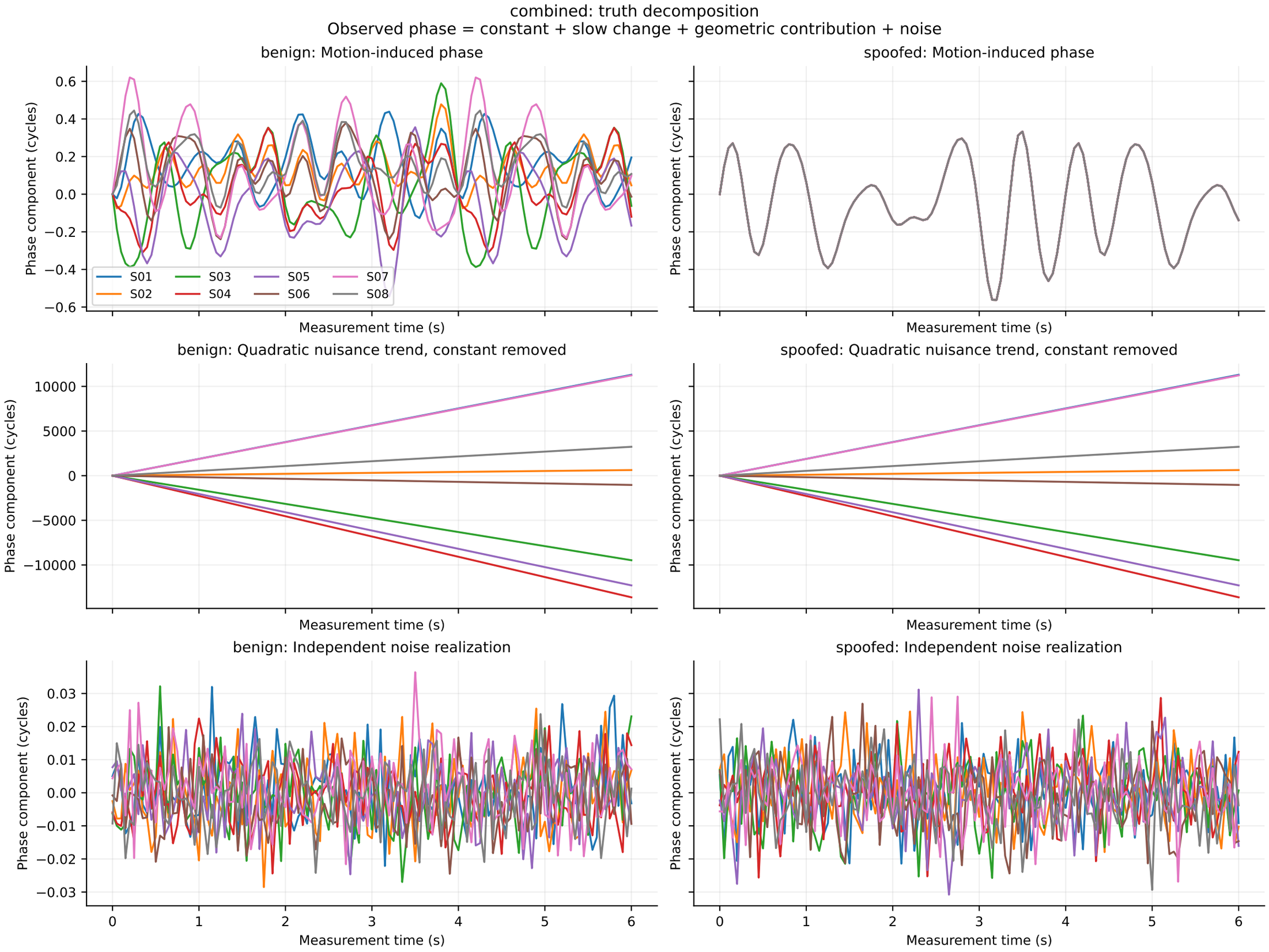

rotation_only: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization
Phase component (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08

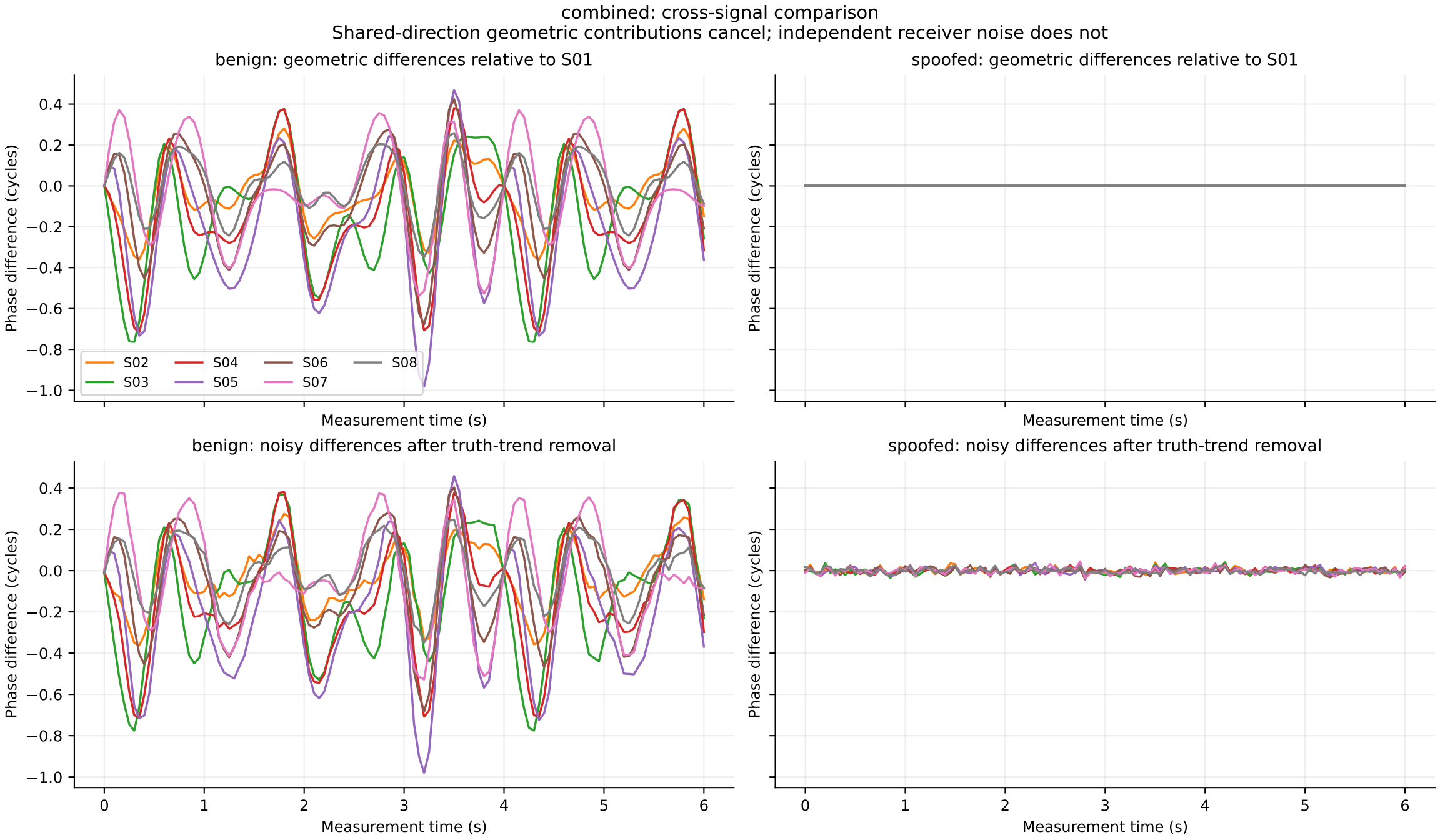

rotation_only: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
Phase difference (cycles)
Measurement time (s)
S02
S03
S04
S05
S06
S07
S08

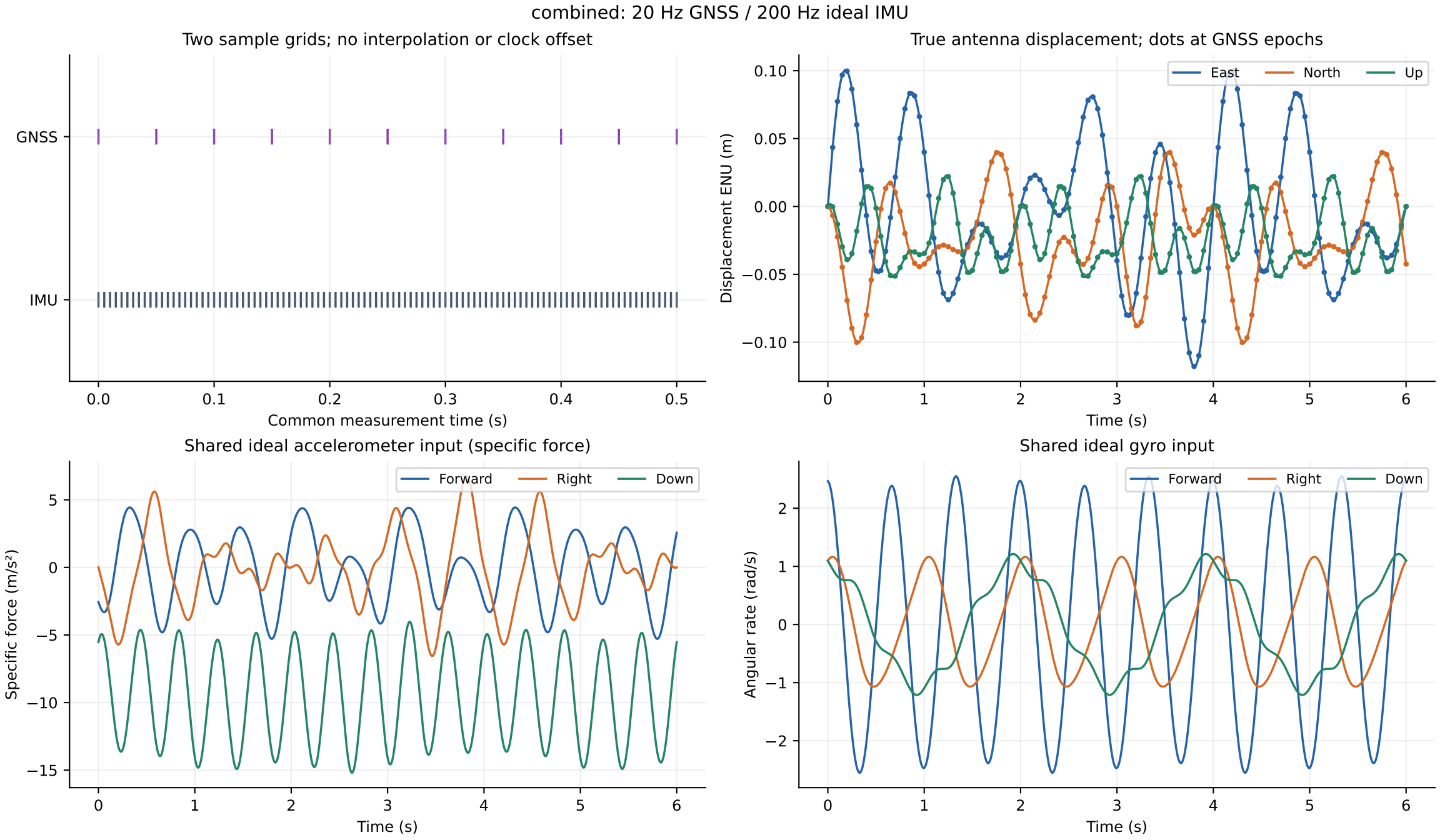

rotation_only: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
2
0
−2
−4
−6
−8
−10
0
1
2
3
4
5
6
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
2
1
0
−1
−2
0
1
2
3
4
5
6
Time (s)

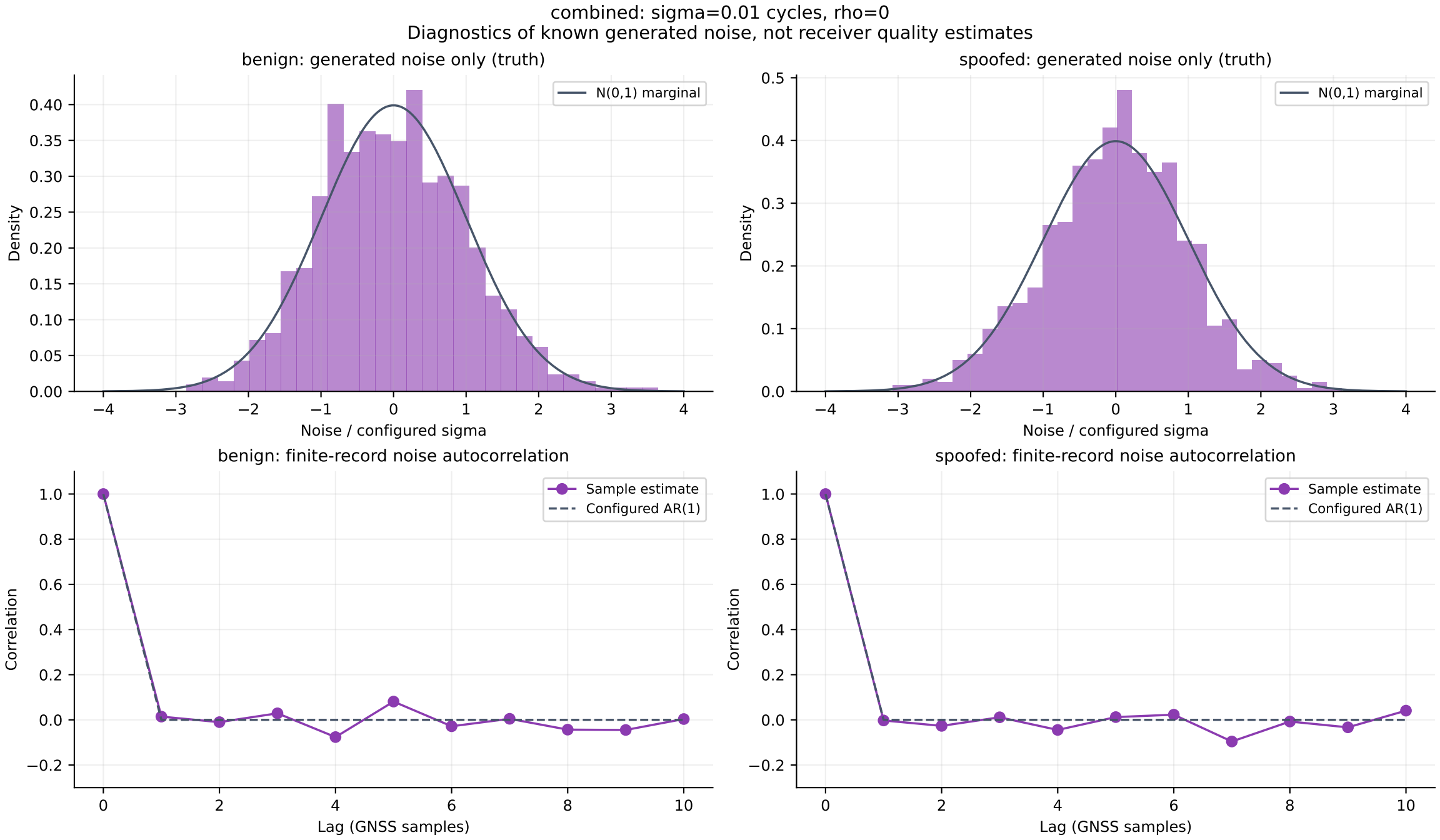

rotation_only: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)


combined: paired observations at 20 Hz
Bottom row uses simulation truth for visualization; no trend estimator or detector is running
benign: observed phase relative to its first sample
spoofed: observed phase relative to its first sample
S01
S02
S03
S04
S05
S06
S07
S08
Observed Δphase (cycles)
10000
5000
0
−5000
−10000
Measurement time (s)
benign: observed phase − known generated trend
spoofed: observed phase − known generated trend
Motion + noise (cycles)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
1
2
3
4
5
6
Measurement time (s)


combined: truth decomposition
Observed phase = constant + slow change + geometric contribution + noise
benign: Motion-induced phase
spoofed: Motion-induced phase
Phase component (cycles)
Measurement time (s)
S01
S02
S03
S04
S05
S06
S07
S08
benign: Quadratic nuisance trend, constant removed
spoofed: Quadratic nuisance trend, constant removed
benign: Independent noise realization
spoofed: Independent noise realization


combined: cross-signal comparison
Shared-direction geometric contributions cancel; independent receiver noise does not
benign: geometric differences relative to S01
spoofed: geometric differences relative to S01
benign: noisy differences after truth-trend removal
spoofed: noisy differences after truth-trend removal
Phase difference (cycles)
Measurement time (s)
S02
S03
S04
S05
S06
S07
S08


combined: 20 Hz GNSS / 200 Hz ideal IMU
Two sample grids; no interpolation or clock offset
GNSS
IMU
0.0
0.1
0.2
0.3
0.4
0.5
Common measurement time (s)
True antenna displacement; dots at GNSS epochs
East
North
Up
Displacement ENU (m)
0.10
0.05
0.00
−0.05
−0.10
0
1
2
3
4
5
6
Time (s)
Shared ideal accelerometer input (specific force)
Forward
Right
Down
Specific force (m/s²)
5
0
−5
−10
−15
0
1
2
3
4
5
6
Time (s)
Shared ideal gyro input
Forward
Right
Down
Angular rate (rad/s)
2
1
0
−1
−2
0
1
2
3
4
5
6
Time (s)


combined: sigma=0.01 cycles, rho=0
Diagnostics of known generated noise, not receiver quality estimates
benign: generated noise only (truth)
spoofed: generated noise only (truth)
N(0,1) marginal
Density
Noise / configured sigma
benign: finite-record noise autocorrelation
spoofed: finite-record noise autocorrelation
Sample estimate
Configured AR(1)
Correlation
Lag (GNSS samples)